K-FIELD SPACECRAFT DYNAMICS PROGRAM

# Reaction-Wheel-Associated Periodic Drift Terms in Spacecraft Tracking and Orbit Determination

K-Field literature synthesis, orbital-harmonic interpretation, and archival test framework

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This report examines published spacecraft tracking, orbit-determination, attitude-control, and reaction-wheel literature for periodic drift terms that require regular compensation. It then defines the observational signatures needed to distinguish conventional wheel disturbances, thruster momentum unloading, environmental forces, estimation artifacts, and a K-Field lateral-acceleration component associated with signed rotational state and inertial orientation.

The approximately twelve-hour interpretation is constrained to cases in which that period has a defensible orbital origin: a one-per-revolution fundamental near twelve hours, as in GPS or GLONASS medium Earth orbit, or a twelve-hour harmonic of a longer orbit, most notably the two-per-revolution harmonic of a geosynchronous sidereal-day orbit. A commanded desaturation schedule or an estimator update interval at twelve hours is not an orbital harmonic and is treated separately.

Mission-specific statements are tied to the source ledger. Derived quantities are marked as calculations from printed inputs. Proposed K-Field regressors are presented as testable model terms rather than assigned retrospectively to residuals without the required telemetry-level fit.

# Document control and scope

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| Title | Reaction-Wheel-Associated Periodic Drift Terms in Spacecraft Tracking and Orbit Determination |
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| Scientific boundary | Published observations, derived mappings, inferred mechanisms, and proposed K-Field tests are kept distinct |


# Abstract

Spacecraft that use reaction wheels or control-moment gyroscopes contain substantial, continuously changing angular momentum. The central question is whether precision navigation records contain periodic translational drift terms correlated with those rotating states and requiring regular compensation. The known literature contains three physically distinct classes of wheel-associated effects. First, wheel-speed harmonics excite structural vibration, line-of-sight jitter, and motion of antenna or instrument reference points. Second, thrusters used to unload stored wheel momentum impart real spacecraft velocity changes, commonly at millimeter-per-second scale. Third, orbit-determination systems estimate empirical constant, once-per-revolution, twice-per-revolution, and pseudo-stochastic accelerations that absorb incompletely modeled solar-radiation pressure, thermal recoil, attitude-dependent forces, small maneuvers, and other deficiencies. These classes can overlap in time but must not be conflated.

The strongest direct published correlation between reaction-wheel speed and a precision tracking disturbance is found in the Gravity Recovery and Interior Laboratory mission. High-frequency Ka-band range-rate disturbances appeared when particular wheels on both spacecraft approached approximately 100 hertz, lasted tens of seconds, and were removed from the gravity solution. Their abrupt speed-localized character is consistent with mechanical resonance and reference-point motion. The same GRAIL processing estimated constant and once-per-orbit empirical accelerations in along-track and cross-track directions, with acceleration prior scales of 10^-9 to 5 x 10^-9 meters per second squared. Those low-frequency solve-for terms are structurally capable of absorbing a smooth inertially referenced acceleration.

Mars Odyssey, RadioAstron, Tianwen-1, and BepiColombo literature establishes that reaction-wheel momentum unloading can dominate predicted trajectory error. Reported unloading velocity increments are typically several millimeters per second. Some operations used a twelve-hour cadence, but those intervals were commanded schedules rather than orbital fundamentals. GNSS orbit solutions provide a separate and more relevant approximately twelve-hour context: GPS and GLONASS orbital periods lie near twelve hours, while a geosynchronous orbit has a sidereal-day fundamental and an approximately 11 h 58 min two-per-revolution harmonic. Empirical GNSS harmonics and pseudo-stochastic parameters therefore form natural absorption channels, but a twelve-hour spectral line alone cannot identify its cause.

No located publication performs the decisive K-Field test: a joint fit of translational tracking data against every wheel axis transformed to inertial space, each wheel inertia, signed wheel speed, wheel acceleration, thruster event history, environmental force geometry, and conventional wheel-angle harmonics. This report develops that fit, identifies parity and phase tests, explains how standard orbit estimation may remove the signal, and ranks archival data sets. GRAIL is the strongest complete public archival experiment. GNSS is the strongest orbital-harmonic experiment because the approximately twelve-hour band is physically tied to the orbit rather than merely to operations.


**Central result**
The literature documents periodic wheel-associated tracking disturbances and regularly compensated trajectory changes, but it does not isolate a continuous center-of-mass acceleration that is odd in signed wheel speed and phase-locked to a fixed inertial K-Field direction. The required telemetry-level regression remains directly executable.

 

# Purpose and evidence layers

This document is a foundational educational reference for spacecraft dynamics, signal analysis, attitude determination, orbit estimation, vibration control, and K-Field dynamics. It begins with the conservation-law structure of reaction-wheel systems, then develops the orbital-frequency mapping needed to interpret periodic residuals. Mission cases are treated in a common evidence framework so that a disturbance in a tracking observable is not automatically misclassified as a true center-of-mass acceleration.

| Layer | Meaning | Example |
|---|---|---|
| Published observation | Quantity or behavior directly described in a cited mission source. | GRAIL Ka-band range-rate disturbances near a wheel-speed region of approximately 100 Hz. |
| Published processing choice | Solve-for parameter, deletion rule, maneuver model, or estimation cadence. | GRAIL quarter-orbit empirical updates; former CODE 12-hour spacing. |
| Derived orbital mapping | Calculation from a published period or geometry. | GEO 2/rev = 23 h 56 min 4 s divided by two. |
| Inferred mechanism | Physical interpretation supported by signature shape but not independently proved. | Mechanical resonance as the economical reading of a narrow speed band. |
| Proposed K-Field test | New regressor or validation condition. | Odd signed-speed response that reverses with wheel direction at fixed inertial orientation. |

 

# Contents

# Notation and units

| Symbol | Definition | Unit or frame |
|---|---|---|
| I_i | Axial moment of inertia of reaction wheel i. | kg m^2 |
| omega_i | Signed angular velocity of wheel i. | rad/s |
| n_i,B | Unit vector along wheel i spin axis in body frame. | body frame |
| R_I<-B | Rotation mapping body vectors into an inertial frame. | dimensionless |
| H_i^I | Signed inertial angular momentum vector of wheel i. | kg m^2/s |
| r-hat, t-hat, n-hat | Radial, transverse or along-track, and orbit-normal unit vectors. | RTN frame |
| u | Orbital phase or argument of latitude. | rad or deg |
| A | Sinusoidal acceleration amplitude. | m/s^2 |
| T | Sinusoidal period. | s |
| L_K | Native Base Cell length unit. | native length |
| g-hat_K | Adopted inertial K-Field unit direction. | ICRS Cartesian |
| Delta-v | Velocity increment from a maneuver. | m/s |

# 1. Scientific question and literature result

## 1.1 The measurement question

A reaction wheel changes spacecraft attitude by exchanging angular momentum with the spacecraft body. The rotor accelerates in one direction while the bus reacts in the opposite direction. In ideal closed-system mechanics, this internal exchange does not accelerate the total center of mass. Real spacecraft are not perfectly closed: solar radiation pressure, thermal recoil, atmospheric drag, magnetic torques, gravity-gradient torques, propellant motion, leaks, and thruster firings provide external forces or torques. Reaction-wheel speeds therefore contain both commanded attitude-control activity and an integrated record of environmental torque.

A search for K-Field lateral acceleration must determine whether a continuous translational acceleration remains after accounting for structural vibration, reference-point motion, environmental-force correlations, and impulsive momentum unloading. The relevant independent variable is not unsigned revolutions per minute alone. It is each wheel angular-momentum vector, including sign and inertial orientation.

## 1.2 Three documented effect classes

| Class | Physical pathway | Expected signature | Center-of-mass implication |
|---|---|---|---|
| Wheel-speed harmonic disturbance | Rotor imbalance, bearing disturbance, motor ripple, cogging, or structural resonance. | Lines at wheel frequency and related harmonics; resonance peaks; line-of-sight or phase-center motion. | Usually no sustained translation of the complete isolated spacecraft center of mass. |
| Momentum unloading | Thrusters or another external actuator remove stored wheel momentum. | Discrete event synchronized with actuator telemetry; velocity or orbit-element step. | Real center-of-mass velocity change for thruster unloading. |
| Empirical orbit acceleration | Estimator adds constant, harmonic, or pseudo-stochastic acceleration parameters. | Low-frequency coefficients at 1/rev, 2/rev, or fixed estimation intervals. | May represent external acceleration, model deficiency, maneuver error, or candidate term. |

The literature result is not one unexplained twelve-hour acceleration. Mechanical wheel disturbances occupy wheel-frequency and structural-mode bands. Thruster unloading occupies discrete event times and produces real velocity increments. Empirical accelerations occupy orbital and sub-orbital bands selected by the estimator.

 

### 1.3 Mission-level result matrix

| Mission or system | Published term | Period, cadence, or scale | Assessment |
|---|---|---|---|
| GRAIL | High-frequency Ka-band range-rate disturbance near particular wheel speeds; constant and 1/rev empirical accelerations. | Approximately 100 Hz wheel region; tens of seconds; 110 min orbit; 10^-9 to 5 x 10^-9 m/s^2 priors. | Direct high-frequency wheel correlation; smooth low-frequency term could be absorbed. |
| Mars Odyssey | Thruster desaturation was a major trajectory-prediction error source. | 3-4 mm/s; 12 h cadence then 20 h; 1.96 h orbit. | Known impulse; twelve hours is operational, not orbital. |
| RadioAstron | SRP drove orbit and wheel momentum; unloads produced 3-7 mm/s. | One or two unloads/day; residual near 8.8 d orbit period. | Demonstrates environmental confounding. |
| GNSS | Constant, 1/rev, 2/rev, sometimes 4/rev terms and pseudo-stochastic parameters. | GPS and GLONASS fundamentals near 12 h; GEO 2/rev near 12 h. | Primary orbital-harmonic search domain. |
| Tianwen-1 | Wheel off-loading and unknown forces aggregated in longer-arc empirical treatment. | Two-way range-rate RMS 0.18-0.35 mm/s. | No separation of steady wheel spin from unloading. |
| BepiColombo study | Desaturation maneuvers treated as solve-for quantities. | Two/day; meter semimajor-axis, approximately 100 m range, cm/s range-rate effects. | Known simulated maneuver sensitivity. |

### 1.4 Missing decisive regression

The reviewed literature does not fit signed wheel angular momentum transformed into inertial space as a translational-force predictor while retaining conventional disturbance and environmental terms. Wheel speed is used to diagnose jitter, estimate torque, schedule unloading, or reconstruct maneuvers. Orbit solutions estimate empirical accelerations without resolving them into per-wheel signed rotational contributions.

**Required distinction**
A periodic residual is not a K-Field detection merely because a reaction wheel was operating. It must follow signed wheel state, reverse under wheel reversal, maintain predicted inertial phase, persist between thruster events, and survive a model including wheel-frequency vibration, wheel acceleration, solar geometry, thermal history, and maneuver impulses.

## 2. Reaction-wheel dynamics and conventional disturbance pathways

### 2.1 Angular-momentum exchange

For a spacecraft body and internal wheels, total angular momentum is the sum of bus and wheel contributions. In the absence of external torque, that total is constant. When a motor changes rotor speed, the bus reacts so angular momentum is redistributed internally.

```
H_total = H_bus + sum_i H_i
H_i^I(t) = R_I<-B(t) [ I_i omega_i(t) n_i,B ]
dH_total/dt = tau_external
```

The body-to-inertial transformation is indispensable because a body-fixed wheel axis may sweep through inertial space as the spacecraft tracks Earth, the Sun, a star, a planet, or local vertical. A scalar speed record cannot encode this geometry.

### 2.2 Center-of-mass null model

For an isolated collection of masses, internal forces occur in action-reaction pairs. Summing component equations eliminates them, leaving only external force on the total center of mass. Rotor acceleration can produce local bearing forces, bus vibration, elastic deformation, and attitude motion but no net force on the complete closed system under the conventional model.

```
M_total a_CM = sum F_external
sum F_internal = 0 for a closed mechanical model
```

This is the null model. The K-Field model adds an external-field coupling dependent on rotational state and orientation; the empirical test is a controlled fit against precision data.

 

## 2.3 Static and dynamic imbalance

A rotor center-of-mass offset produces a rotating bearing force, while principal-axis misalignment produces a rotating moment. To first order, the imbalance-force envelope grows with angular speed squared and the direction rotates with wheel angle. Dominant spectral content lies at the wheel fundamental and harmonics, not the slow orbital period.

```
F_imbalance(t) approximately m_e e omega^2 [cos(theta_w), sin(theta_w)]
theta_w(t) = integral omega(t) dt + theta_w(0)
```

The square dependence is even under reversal. A minimal signed-speed K-Field term is odd. This parity difference is a strong diagnostic.

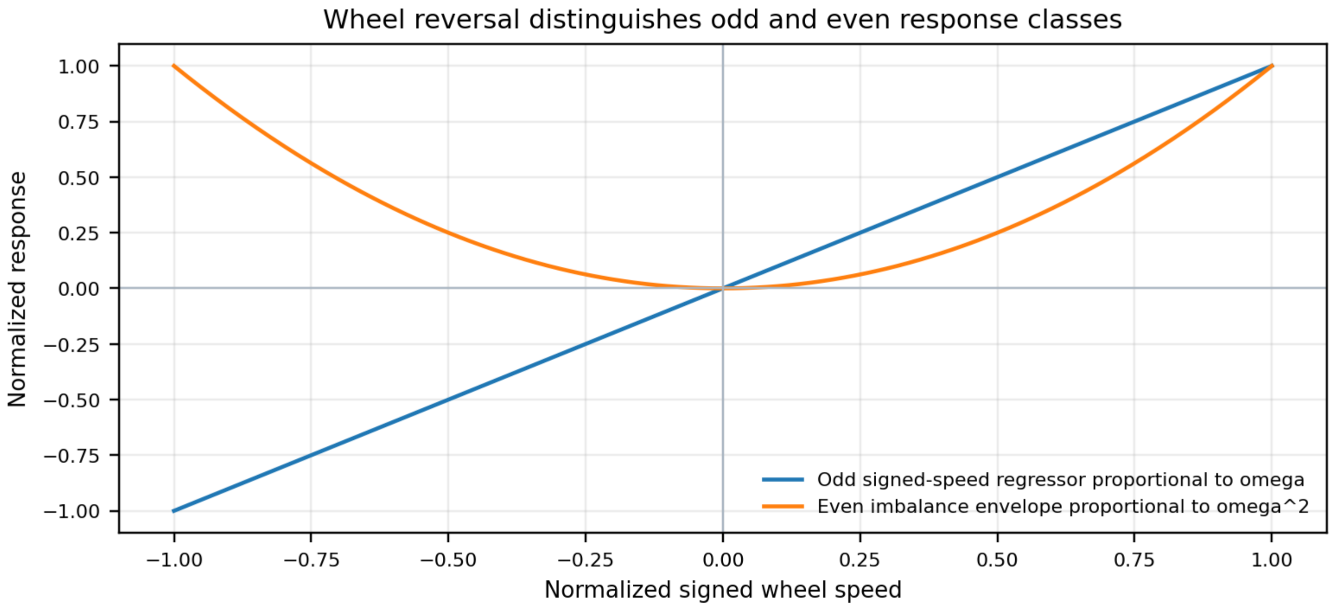


**Figure 1. Odd signed-speed and even imbalance response.** The linear signed-speed curve changes sign under wheel reversal. The quadratic imbalance envelope does not. Mechanical disturbances can contain additional terms, but reversal parity remains central.

## 2.4 Bearings, motor ripple, cogging, and resonance

Reaction-wheel assemblies produce disturbance spectra through bearing imperfections, lubricant behavior, motor commutation, electromagnetic torque ripple, cogging, rotor imbalance, and assembly modes. A forcing line can intersect a flexible spacecraft mode as wheel speed changes, producing a narrow speed band with large response. The observation depends on the transfer path to the antenna or instrument reference point. A phase-center displacement can appear as range-rate error without center-of-mass motion. Dennehy surveys these wheel harmonics and structural pathways [R6].

## 2.5 Wheel acceleration and control activity

Commanded motor torque is proportional to the rate of wheel angular momentum. Motor current, wheel acceleration, and attitude-controller error signals form a separate predictor family. A transient correlated with wheel acceleration can arise from bus reaction torque, elastic deformation, electronics coupling, or control-loop interaction and must be separated from a steady signed-speed term.

```
q_dotomega,i(t) = I_i d omega_i/dt
q_omega,i(t) = I_i omega_i(t)
```

## 2.6 Momentum accumulation and unloading

External torque gradually changes wheel momentum. Near an operational limit, an external actuator transfers angular momentum out of the wheel system. Chemical thrusters produce direct translational impulse in addition to torque. The maneuver changes wheel speeds, but the trajectory change is caused by the external actuator.

```
Delta-v_k = integral over event k of a_thruster(t) dt
a_unload(t) approximately sum_k Delta-v_k delta(t - t_k)
```

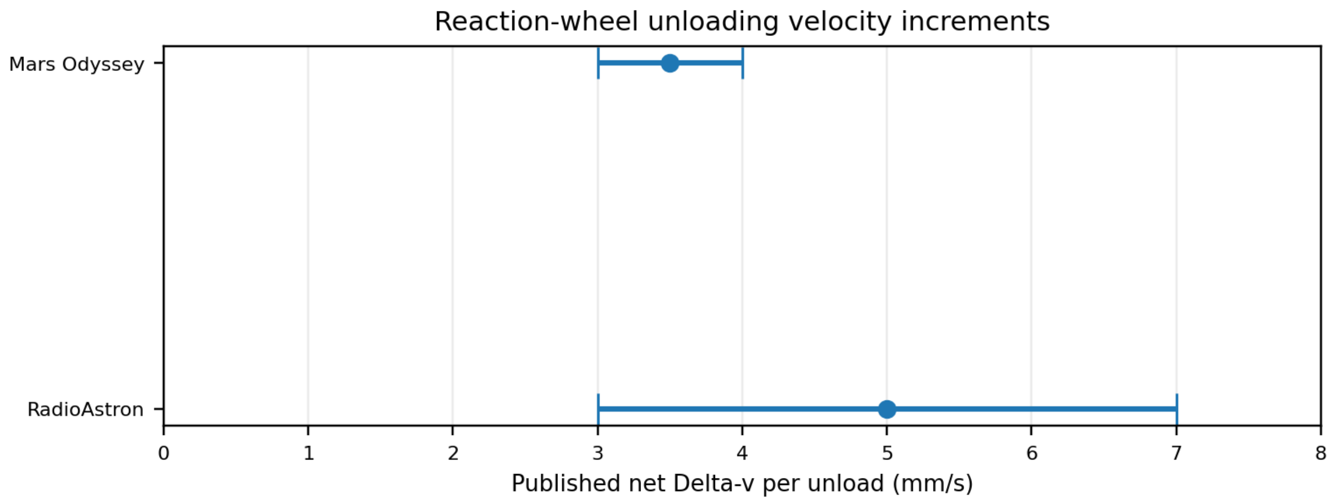


**Figure 2. Published unloading velocity increments.** Mars Odyssey reports typical desaturation increments of 3-4 mm/s. RadioAstron reports 3-7 mm/s. Intervals are direct source ranges, not confidence intervals.

### 2.7 Environmental torque as a confounder

Solar radiation pressure can accelerate the center of mass and exert torque. Translation enters the orbit directly; torque is integrated by the attitude-control system and appears as wheel-momentum change. Wheel speed and orbit residual can therefore correlate even when the wheel is only a sensor of external torque. RadioAstron is a clear example: wheel histories constrained SRP torque while SRP force and unloading separately affected the orbit [R4-R5].

## 3. Orbital frequencies and the corrected approximately twelve-hour interpretation

### 3.1 Four different meanings of a period

A repeated term may be indexed by wheel angle, attitude cycle, orbit revolution, Earth rotation, thermal cycle, maneuver schedule, tracking schedule, or estimator knot spacing. Equal numerical periods do not imply equal mechanisms. The first task is to identify what phase variable closes the signal.

| Period class | Phase variable | Example | Discriminator |
|---|---|---|---|
| Wheel harmonic | Rotor angle. | Imbalance at wheel frequency. | Moves with wheel speed and tracks integrated wheel angle. |
| Orbital harmonic | Orbital phase. | 1/rev or 2/rev acceleration. | Maintains phase relative to orbit geometry. |
| Attitude cycle | Commanded orientation. | Yaw-steering thermal or SRP effects. | Tracks quaternions and illumination. |
| Operational cadence | Commanded event times. | Desaturation every 12 or 20 h. | Changes when operations change. |
| Estimator cadence | Preselected knots or arc boundaries. | Parameters every 12 h. | Exists in model architecture without a physical oscillator. |

### 3.2 Geosynchronous orbit

A geosynchronous orbit repeats with Earth sidereal rotation, approximately 23 hours 56 minutes 4 seconds. Its fundamental is near one sidereal day, not twelve hours. A term near 11 hours 58 minutes 2 seconds is its second orbital harmonic, 2/rev. It can arise from twofold symmetry, a quadratic direction-cosine term, or an explicit twice-per-orbit estimator basis.

```
T_GEO = 23 h 56 min 4 s = 86164 s
T_GEO,2/rev = T_GEO / 2 = 11 h 58 min 2 s
```

 

Geostationary orbit is the circular, equatorial, prograde subset of geosynchronous orbit. The period statement remains a sidereal-day fundamental.

## 3.3 GPS, GLONASS, and Galileo

GPS satellites circle Earth approximately twice per sidereal day, giving a 1/rev period near 11 h 58 min. GLONASS uses 11 h 15 min 44 s. These are genuine approximately twelve-hour fundamentals. Galileo uses 14 h 4 min 45 s and should not be forced into a twelve-hour category. For Galileo, 2/rev is near 7.04 h.

## 3.4 Other missions

GRAIL orbited the Moon in approximately 110 min. Mars Odyssey orbited Mars in approximately 1.96 h. RadioAstron residual analysis identified a multi-day component near an approximately 8.8-day orbit. Their orbital harmonics occupy distinct bands from twelve hours.

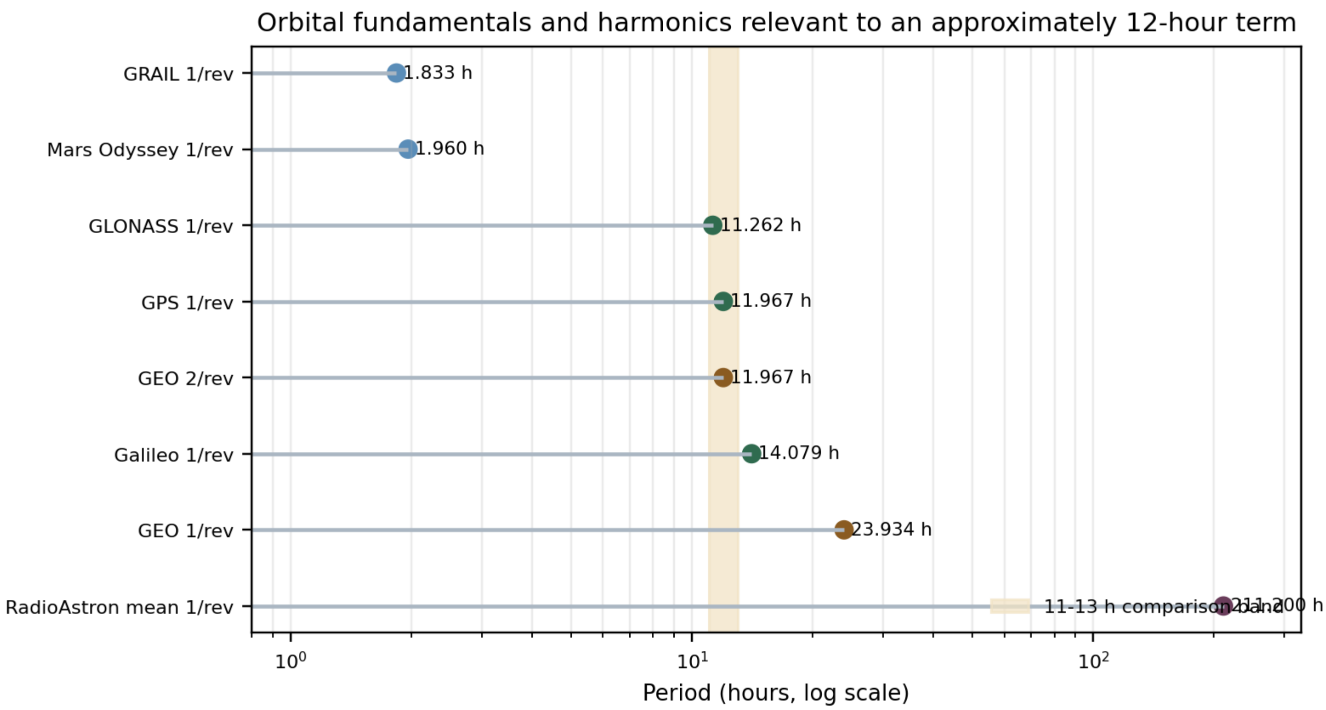


**Figure 3. Orbital fundamentals and harmonics near the twelve-hour band.** The 11-13 h band contains GPS and GLONASS 1/rev periods and GEO 2/rev. GRAIL, Odyssey, Galileo, GEO 1/rev, and RadioAstron occupy different bands.

| System or term | Period (h) | Classification | 12 h relevance | Source |
|---|---|---|---|---|
| GRAIL 1/rev | 1.833333333 | orbital fundamental | No direct 12 h identification | R1 |
| Mars Odyssey 1/rev | 1.960000000 | orbital fundamental | No direct 12 h identification | R3 |
| GLONASS 1/rev | 11.262222222 | orbital fundamental | Direct orbital relevance | R13 |
| GPS 1/rev | 11.967222222 | orbital fundamental | Direct orbital relevance | R11 |
| GEO 2/rev | 11.967222222 | second orbital harmonic | Direct orbital relevance | R12 |
| Galileo 1/rev | 14.079166667 | orbital fundamental | No direct 12 h identification | R14 |
| GEO 1/rev | 23.934444444 | orbital fundamental | Parent fundamental of GEO 2/rev | R12 |
| RadioAstron mean 1/rev | 211.200000000 | approximate orbital fundamental | No direct 12 h identification | R5 |

## 3.5 Fixed inertial acceleration in RTN

For a circular orbit, let p and q span the orbit plane and h be normal. A fixed inertial acceleration projects as sine and cosine at one cycle per revolution in radial and along-track components, while the normal projection is constant in the ideal fixed-plane case.

```
r_hat(u) = p cos(u) + q sin(u)
t_hat(u) = -p sin(u) + q cos(u)
n_hat = h
a_R = a_p cos(u) + a_q sin(u)
a_T = -a_p sin(u) + a_q cos(u)
a_N = a_h
```

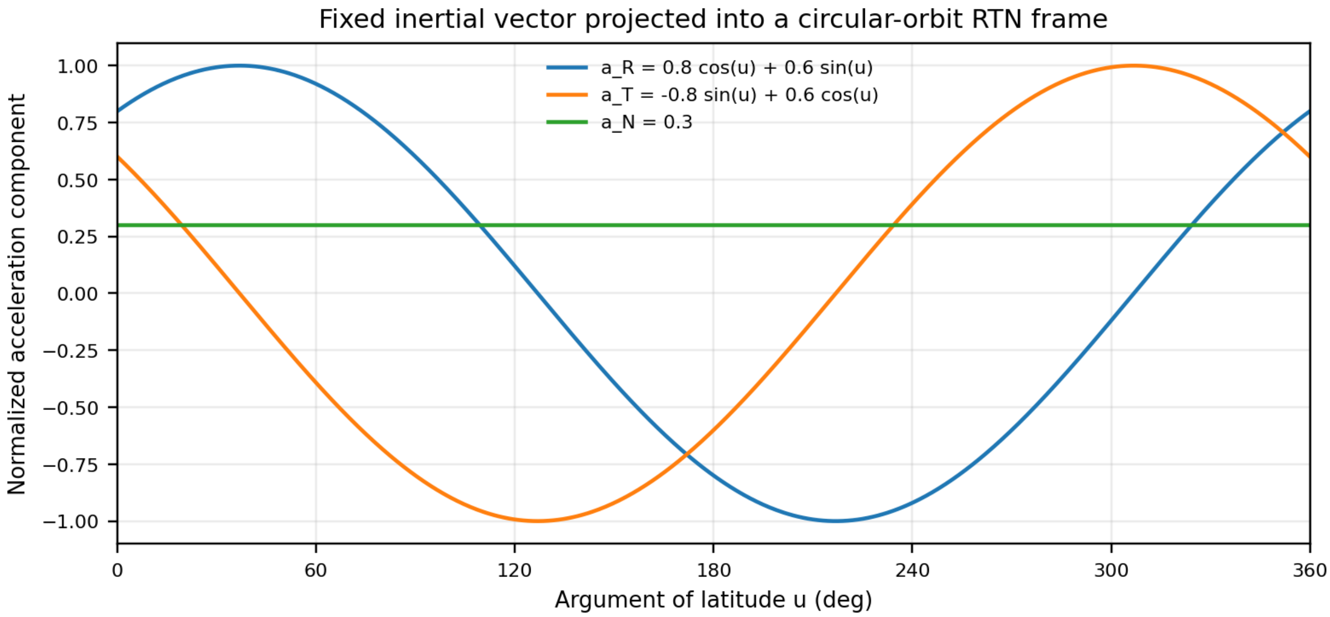


**Figure 4. Projection of a fixed inertial vector into RTN.** The normalized example uses a_p = 0.8, a_q = 0.6, and a_h = 0.3. Radial and along-track terms are 1/rev; the normal term is constant.

This mapping explains why a fixed K-Field acceleration may be estimated by standard 1/rev coefficients and why its period depends on the spacecraft orbit.

## 3.6 Second harmonic and twofold symmetry

A linear projection produces 1/rev. A 2/rev term arises from a quadratic projection, an unsigned magnitude, a two-lobed geometry, or a product of two 1/rev terms.

```
cos^2(u) = [1 + cos(2u)] / 2
sin(u) cos(u) = sin(2u) / 2
```

This is the relevant route from a sidereal-day GEO orbit to an approximately twelve-hour term. The model must specify the symmetry or nonlinear projection that doubles the frequency.

## 3.7 Scheduled events and estimator spacing

Mars Odyssey changed desaturation cadence from twelve to twenty hours while its orbit remained approximately 1.96 h. BepiColombo studies assumed two desaturations per day. These recurrences are command-driven. Former CODE processing used twelve-hour pseudo-stochastic spacing; later rescheduling reduced orbit misclosures [R7]. That value is an estimator bandwidth, not a detected force.

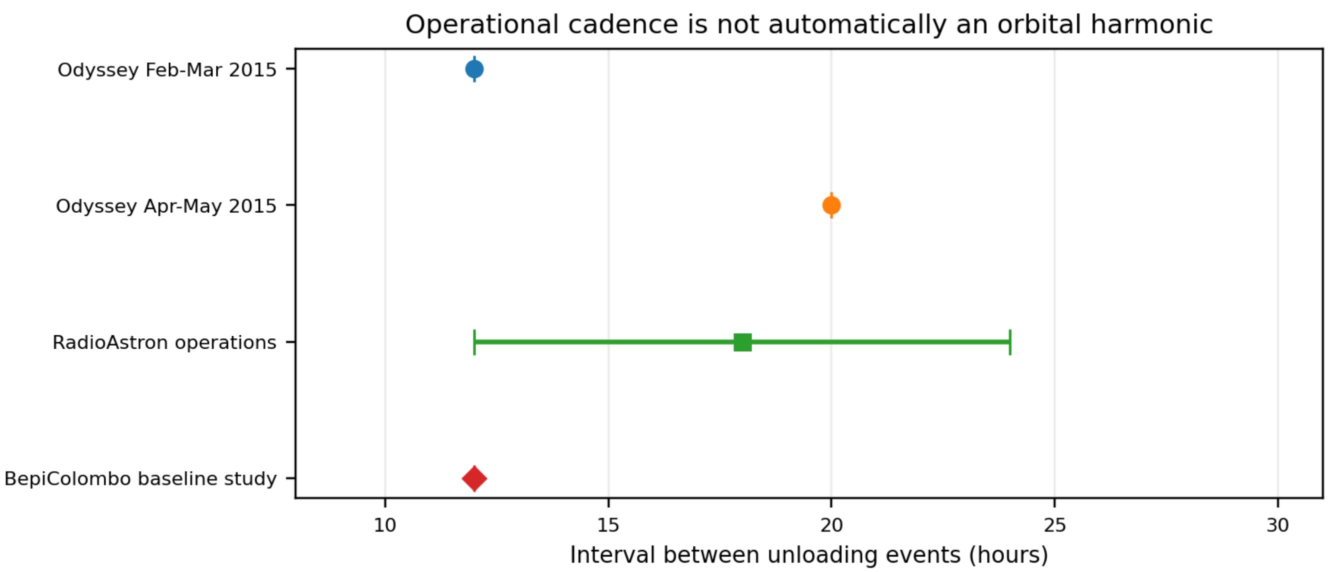


**Figure 5. Operational momentum-unloading cadence.** Published or study-defined intervals are shown. The same 12 h number appears in different operational contexts and is not thereby orbital.

> **Twelve-hour rule**
> Interpret a roughly twelve-hour term as orbital only after demonstrating phase closure with a near-twelve-hour 1/rev orbit, a defined 2/rev symmetry of a near-one-day orbit, or another explicitly derived orbital combination. Command schedules and estimator knots are separate classes.

# 4. K-Field Base Cell, planes, and rotational coupling geometry

## 4.1 Role of the Base Cell

The literature search identifies where a rotationally coupled acceleration could appear. The K-Field Base Cell supplies the locked geometry used to define field-direction and projection terms in the broader program. It is a right-handed parallelepiped generated by vectors a, b, and c in native length unit L_K, with nodes O, A, B, C, AB, AC, BC, and ABC and generating planes P_ab, P_ac, and P_bc.

## 4.2 Locked scalar geometry

| Quantity | Value | Unit |
|---|---|---|
| \|a\| | 0.0207970587 | L_K |
| \|b\| | 0.0364942052 | L_K |
| \|c\| | 0.0493085660 | L_K |
| angle(a,b) | 83.933483 | deg |
| angle(a,c) | 80.448130 | deg |
| angle(b,c) | 50.363232 | deg |
| A_ab | 0.000754721788097 | L_K^2 |
| A_ac | 0.001011255755679 | L_K^2 |
| A_bc | 0.001385784440910 | L_K^2 |
| Volume | 2.842066945211217e-05 | L_K^3 |

 

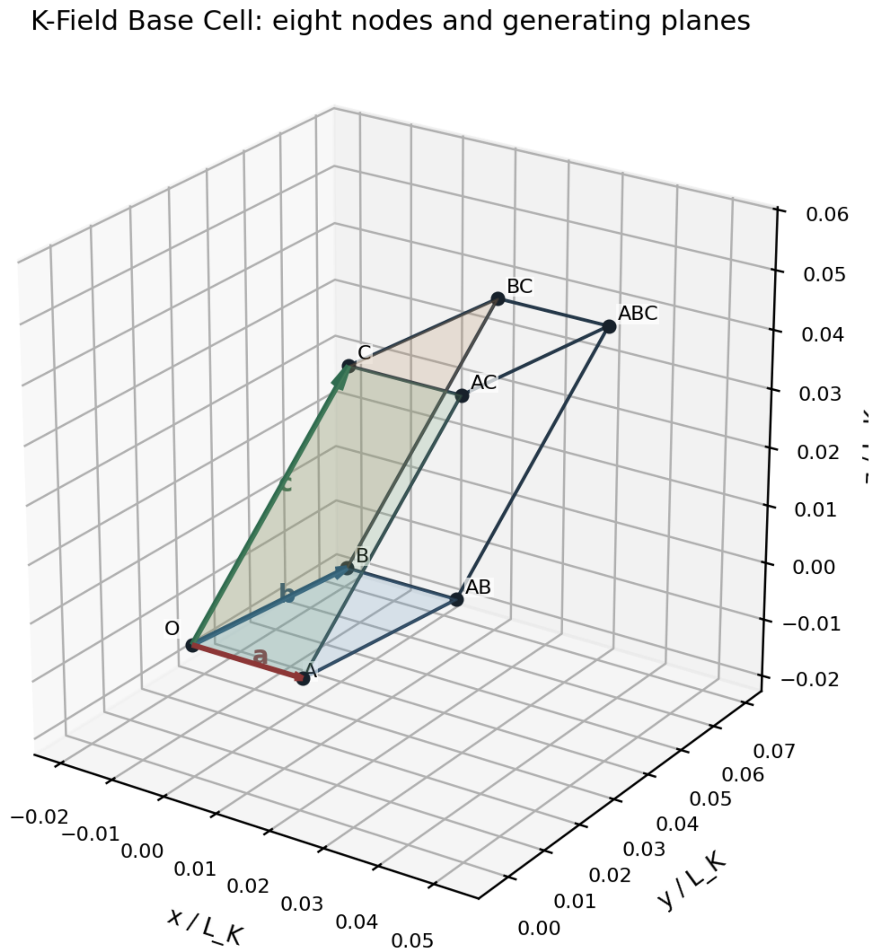


**Figure 6. K-Field Base Cell: eight nodes and generating faces.** Constructed directly from locked edge magnitudes and pairwise angles. All eight vertices are labeled; coordinates and plane equations appear below and in Appendix B.

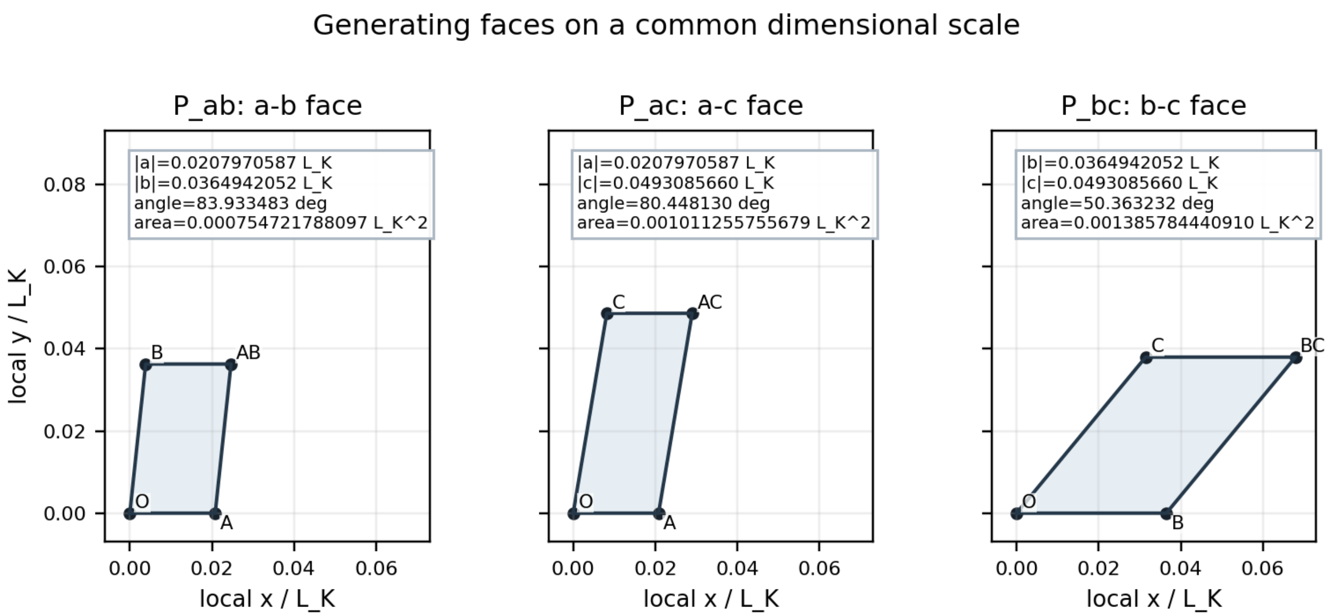


**Figure 7. Generating planes on a common dimensional scale.** The a-b, a-c, and b-c parallelograms are drawn without independent rescaling. Exact lengths, angles, and areas are printed above each face and every vertex is labeled.

## 4.3 Right-handed coordinate realization

The coordinate realization fixes a along positive x, places b in the x-y plane with positive y, and chooses the positive-z branch for c. This preserves all locked lengths and angles with positive scalar triple product.

| Node | x / L_K | y / L_K | z / L_K | Definition |
|---|---|---|---|---|
| O | 0.000000000000 | 0.000000000000 | 0.000000000000 | origin |
| A | 0.020797058700 | 0.000000000000 | 0.000000000000 | a |
| B | 0.003856816120 | 0.036289833047 | 0.000000000000 | b |
| C | 0.008182284384 | 0.030762382380 | 0.037657147177 | c |
| AB | 0.024653874820 | 0.036289833047 | 0.000000000000 | a + b |
| AC | 0.028979343084 | 0.030762382380 | 0.037657147177 | a + c |
| BC | 0.012039100504 | 0.067052215428 | 0.037657147177 | b + c |
| ABC | 0.032836159204 | 0.067052215428 | 0.037657147177 | a + b + c |

## 4.4 Face planes and normals

Each generating plane passes through O and is spanned by two edges. The unit normal is the normalized cross product in the listed order. Reversing a normal changes sign but not the plane; the opposite-face offset changes consistently.

| Plane | Spanning vectors | Unit normal (x, y, z) | Origin face | Opposite-face offset |
|---|---|---|---|---|
| P_ab | a, b | (0.000000000000, 0.000000000000, 1.000000000000) | n_ab dot x = 0 | 0.037657147177 |
| P_ac | a, c | (0.000000000000, -0.774440981837, 0.632646161492) | n_ac dot x = 0 | -0.028104333936 |
| P_bc | b, c | (0.986135753693, -0.104804678115, -0.128655566277) | n_bc dot x = 0 | 0.020508723156 |

## 4.5 Inertial field direction and minimal regressor

The adopted normalized K-Field direction is expressed in ICRS Cartesian components. Its sign convention remains fixed across all mission arcs.

```
g_hat_K = (-0.945636157362, -0.154606293310, 0.286127859460)
|g_hat_K| = 1
```

For wheel axis n_i^I, the minimal lateral orientation factor is n_i^I x g_hat_K. Its magnitude is maximal for orthogonal axes and zero for parallel axes.

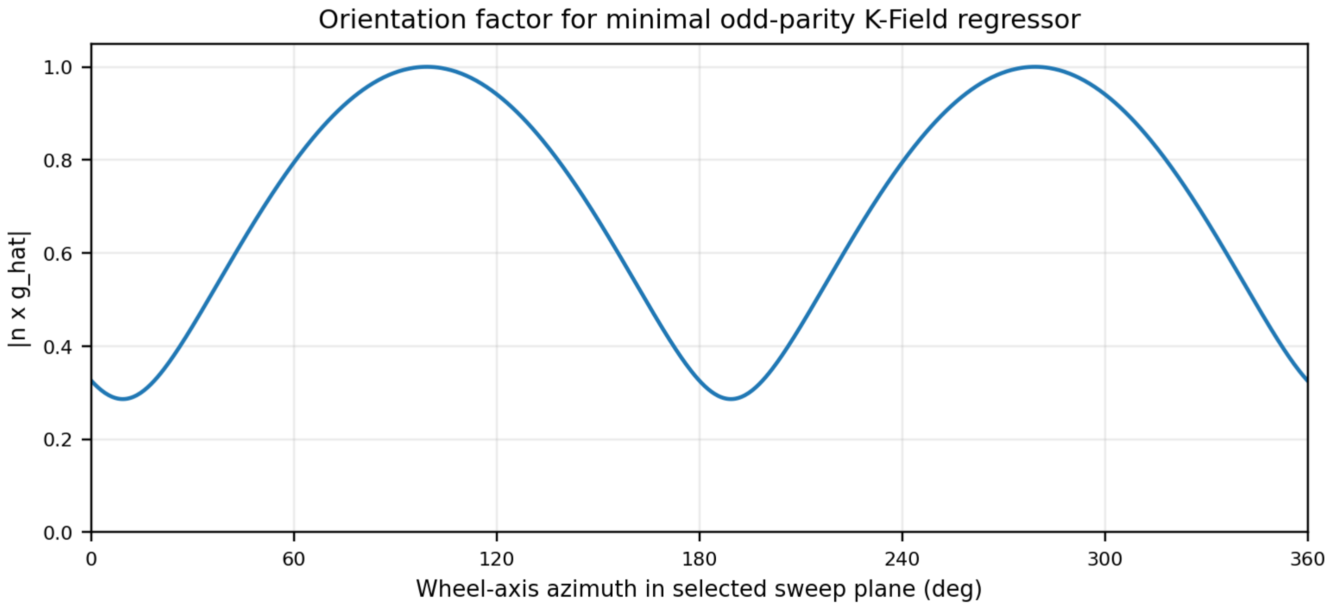


**Figure 8. Orientation factor for a representative axis sweep.** Computed from |n x g_hat_K| for unit axes swept in a selected plane. A spacecraft envelope depends on actual wheel layout and attitude.

### 4.6 Per-wheel and system-level state

```
H_i^I(t) = R_I<-B(t) I_i omega_i(t) n_i,B
H_RW^I(t) = sum_i H_i^I(t)
q_K^I(t) = sum_i I_i omega_i(t) [n_i^I(t) x g_hat_K]
a_K^I(t) = alpha_K q_K^I(t)
```

The minimal proxy tests the defining sign and inertial-orientation structure without unnecessary degrees of freedom. The units of alpha_K depend on the selected normalization and must be reported.

### 4.7 Predicted discriminators

| Property | K-Field signed-speed prediction | Conventional comparator |
|---|---|---|
| Wheel reversal | Contribution changes sign after axis convention. | Imbalance envelope is primarily even in speed. |
| Steady speed | Candidate can remain nonzero between transients. | Motor-torque and d omega/dt terms approach zero. |
| Inertial attitude change | Direction follows n_i^I relative to g_hat_K. | SRP/thermal follow Sun and surface geometry; structural response follows body geometry. |
| Resonance crossing | No narrow mechanical resonance required by minimal model. | Mechanical response can peak sharply at a structural mode. |
| Thruster unloading | Continuous predictor changes with new wheel state but has no impulse. | Thruster telemetry identifies direct Delta-v. |

## 5. Orbit determination, periodic solve-for terms, and signal absorption

### 5.1 What orbit determination estimates

Orbit determination combines a force model, initial state, spacecraft properties, maneuver and attitude histories, observation models, and tracking data. Selected parameters are adjusted to minimize weighted residuals. Low post-fit residuals do not prove that every force is correct; empirical parameters can absorb omitted accelerations.

 

## 5.2 Constant and orbital-harmonic accelerations

A common empirical model applies constant plus sine and cosine accelerations in radial, along-track, or cross-track directions. Because a fixed inertial vector projects into RTN at 1/rev, these terms can absorb a continuous inertial acceleration. Quarter-orbit updates can follow slow amplitude changes.

```
a_RTN(u) = C_0 + C_1c cos(u) + C_1s sin(u)
optional: + C_2c cos(2u) + C_2s sin(2u) + ...
```

## 5.3 Pseudo-stochastic accelerations and arcs

Piecewise accelerations or velocity pulses are introduced at selected epochs to absorb force deficiencies. Short spacing increases flexibility but can remove a coherent physical signal. Independent arc initial states similarly absorb long-period position and velocity components. The former twelve-hour CODE spacing is therefore an absorption channel, not proof of a twelve-hour force.

## 5.4 Observation reference points

Radio tracking is referenced to an antenna phase center, laser ranging to a reflector, and inter-spacecraft ranging to instrument-defined points. Attitude motion and elastic deformation change their vectors relative to the center of mass. A wheel-frequency displacement at an antenna can create apparent range-rate disturbance without center-of-mass velocity.

## 5.5 Sinusoidal acceleration scale

For a(t) = A cos(omega t), the particular velocity amplitude is A/omega and displacement amplitude is A/omega squared. These are dimensional sensitivity scales, not guaranteed post-fit residuals, because initial conditions, dynamics, geometry, and empirical parameters can absorb the signal.

```
v_amp = A / omega = A T / (2 pi)
x_amp = A / omega^2 = A T^2 / (4 pi^2)
```

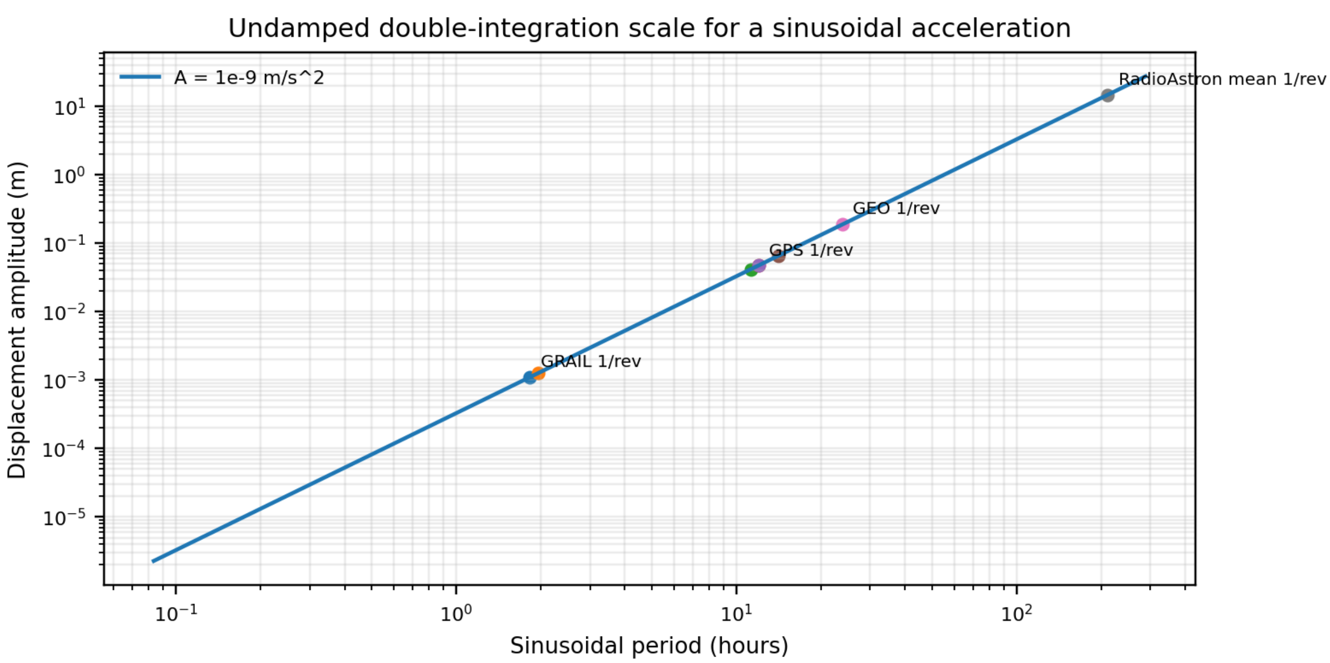


**Figure 9. Displacement scale from sinusoidal acceleration.** Calculated for A = 10^-9 m/s^2. Displacement grows as period squared. Mission points use the period ledger.

 

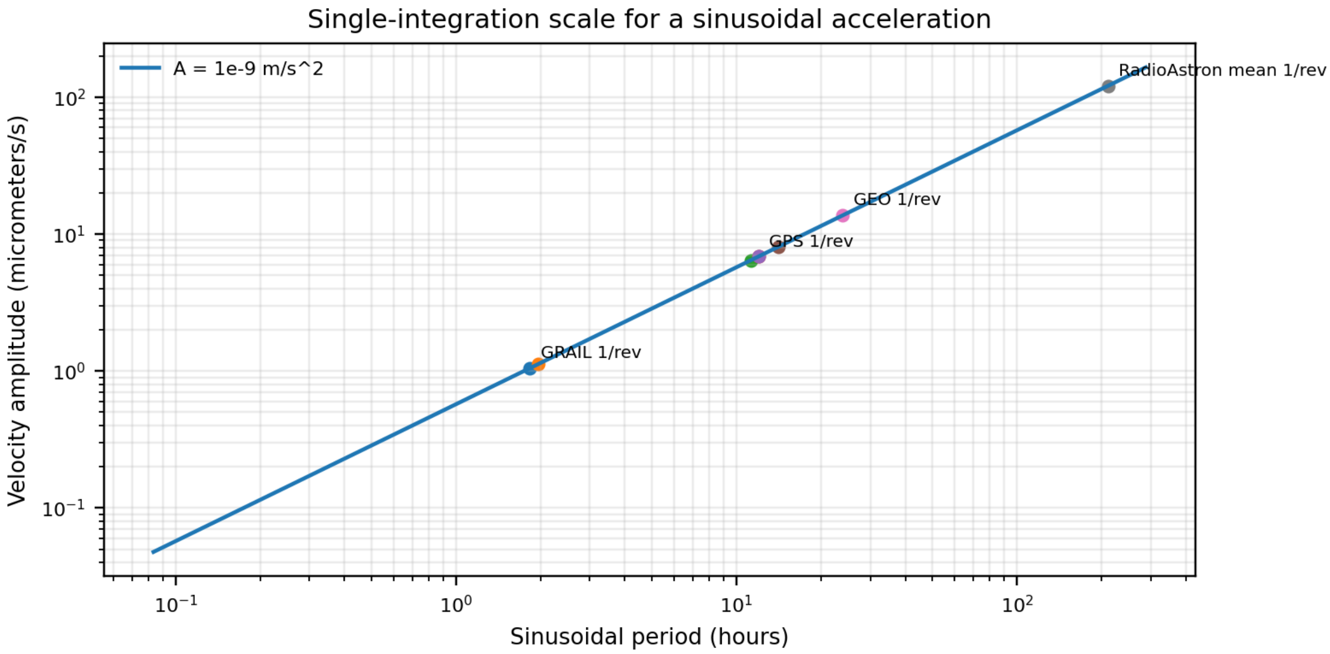


**Figure 10. Velocity scale from sinusoidal acceleration.** Calculated for A = 10^-9 m/s^2. Velocity grows linearly with period. Values precede orbit-fit absorption.

| Term | Period | x amplitude at 1e-9 m/s^2 | v amplitude at 1e-9 m/s^2 | Meaning |
|---|---|---|---|---|
| GRAIL 1/rev | 1.833333 h | 0.00110338769 m | 1.05042262e-06 m/s | Undamped particular-solution scale |
| Mars Odyssey 1/rev | 1.960000 h | 0.00126112289 m | 1.12299728e-06 m/s | Undamped particular-solution scale |
| GLONASS 1/rev | 11.262222 h | 0.0416383441 m | 6.45277801e-06 m/s | Undamped particular-solution scale |
| GPS 1/rev | 11.967222 h | 0.0470145167 m | 6.85671326e-06 m/s | Undamped particular-solution scale |
| GEO 2/rev | 11.967222 h | 0.0470145167 m | 6.85671326e-06 m/s | Undamped particular-solution scale |
| Galileo 1/rev | 14.079167 h | 0.0650727507 m | 8.06676829e-06 m/s | Undamped particular-solution scale |
| GEO 1/rev | 23.934444 h | 0.188058067 m | 1.37134265e-05 m/s | Undamped particular-solution scale |
| RadioAstron mean 1/rev | 211.200000 h | 14.6431022 m | 0.000121008686 m/s | Undamped particular-solution scale |

At exactly twelve hours, 10^-9 m/s^2 gives approximately 0.0473 m displacement and 6.88 micrometers per second velocity. At 110 minutes, it gives approximately 1.10 mm and 1.05 micrometers per second. These scales are compatible with precision-orbit relevance and empirical absorption.

## 5.6 Identifiability and experimental diversity

A K-Field coefficient is identifiable only when its time history is independent of other design-matrix columns. Wheel speed responding to solar torque correlates with SRP. Unloads at fixed orbit phase correlate with impulses. Fixed attitude over short arcs makes orientation nearly constant and correlated with initial velocity or constant acceleration. Diversity in reversals, attitudes, Sun angles, wheel combinations, orbital planes, and spacecraft is the remedy.

## 5.7 Pre-fit, post-fit, and deleted data

Published post-fit residuals may no longer contain the signal if empirical accelerations, maneuver parameters, arc states, or editing removed it. A rigorous search uses pre-fit observations or reconstructs the orbit with a restricted baseline model, then adds candidate terms in a controlled sequence. Known wheel-resonance intervals are a separate mechanical control set.

 

# 6. Literature search method and source qualification

## 6.1 Objective and inclusion criteria

The search targeted periodic or recurrent trajectory and tracking terms associated with reaction-wheel operation, including effects requiring routine compensation, estimation, deletion, or maneuver reconstruction. It covered lunar orbiters, Mars orbiters, highly elliptical Earth orbiters, GNSS precise orbit determination, and Mercury mission simulations.

| Criterion | Requirement |
|---|---|
| Source authority | Peer-reviewed journal, mission conference paper, official archive specification, or official constellation documentation. |
| Wheel relevance | Reaction-wheel disturbance, momentum accumulation, unloading, or wheel-dependent estimation explicitly discussed. |
| Navigation relevance | Effect appears in orbit prediction, range, range rate, Doppler, inter-spacecraft tracking, or empirical acceleration. |
| Numerical content | At least one period, cadence, acceleration, velocity increment, residual, or parameter is available. |
| Mechanism separability | Enough context to distinguish structural disturbance, maneuver impulse, environmental force, or estimator parameterization. |

## 6.2 Exclusions and extraction fields

General summaries and anecdotes were not used for mission-specific numerical claims. No residual was labeled as K-Field evidence solely by resemblance in period. Extracted fields included orbit class, wheel-state type, tracking observable, recurrence, duration, magnitude, processing action, and the telemetry missing from a decisive signed-speed test.

| Field | Purpose |
|---|---|
| Mission and orbit class | Defines physically relevant fundamental and harmonic frequencies. |
| Wheel-state information | Separates signed speed, unsigned speed, momentum, acceleration, and operating status. |
| Tracking observable | Separates trajectory sensitivity from reference-point or line-of-sight disturbance. |
| Recurrence and duration | Distinguishes resonance, orbit harmonic, schedule, and environmental cycle. |
| Magnitude | Compares acceleration, Delta-v, range, range rate, and orbit error. |
| Processing action | Records deletion, empirical fit, maneuver estimation, scaling, or arc changes. |
| K-Field relevance | Identifies missing telemetry and fit structure. |

## 6.3 Traceability

Every mission-specific number used in a table or figure is tied to R1-R16. Derived quantities, such as GEO 2/rev and sinusoidal integration scales, are calculated in the companion JSON from published inputs. Appendices provide the source and metric ledgers.

 

# 7. GRAIL: direct wheel-speed correlation in precision inter-spacecraft tracking

## 7.1 Mission and observable

GRAIL used two nearly identical spacecraft in low lunar orbit. The primary observable was separation change measured by the Ka-band Lunar Gravity Ranging System. Individual spacecraft orbits were supported by Deep Space Network radiometric tracking. The inter-spacecraft link was exceptionally sensitive to differential motion and to displacement of measurement reference points. Goossens and coauthors describe the gravity processing [R1], while the official PDS interface specification defines wheel speed, attitude, thruster, spacecraft geometry, mass properties, radiometric, and ranging products [R2].

## 7.2 Wheel-speed-associated high-frequency events

The strongest direct published wheel correlation found is a class of sudden high-frequency GRAIL Ka-band range-rate disturbances. They frequently occurred when particular reaction wheels on both spacecraft approached approximately 100 Hz. Events lasted tens of seconds and affected intervals were deleted from gravity-field processing [R1].

The signature proves that wheel state can enter a precision translational tracking observable, but its abrupt speed-localized character is consistent with structural or instrument resonance. A smooth center-of-mass acceleration proportional to signed wheel speed would not ordinarily switch on only in a narrow speed band unless the coupling law itself contained a resonance.

| Feature | Published behavior | Conventional reading | K-Field use |
|---|---|---|---|
| Speed localization | Near approximately 100 Hz for particular wheels. | Forcing line intersects a structural or instrument mode. | Mechanical-resonance control; exclude from smooth fit. |
| Duration | Tens of seconds. | Brief passage through or residence in resonant band. | Define event and guard windows. |
| Observable | Ka-band inter-spacecraft range rate. | Reference-point or differential structural motion can enter directly. | Compare with individual DSN tracking. |
| Processing | Affected data deleted. | Known contamination removed before inversion. | Analyze separately as transfer-function control. |

## 7.3 Empirical acceleration model

The GRAIL solution estimated constant and once-per-orbit sine and cosine accelerations in along-track and cross-track directions. Six empirical acceleration parameters per spacecraft were estimated over approximately one-quarter-orbit intervals. Prior standard deviations were approximately 10^-9 m/s^2 in the primary mission and 5 x 10^-9 m/s^2 in the extended mission [R1]. A fixed inertial acceleration maps into 1/rev orbital components, so these terms can absorb a smooth K-Field component.

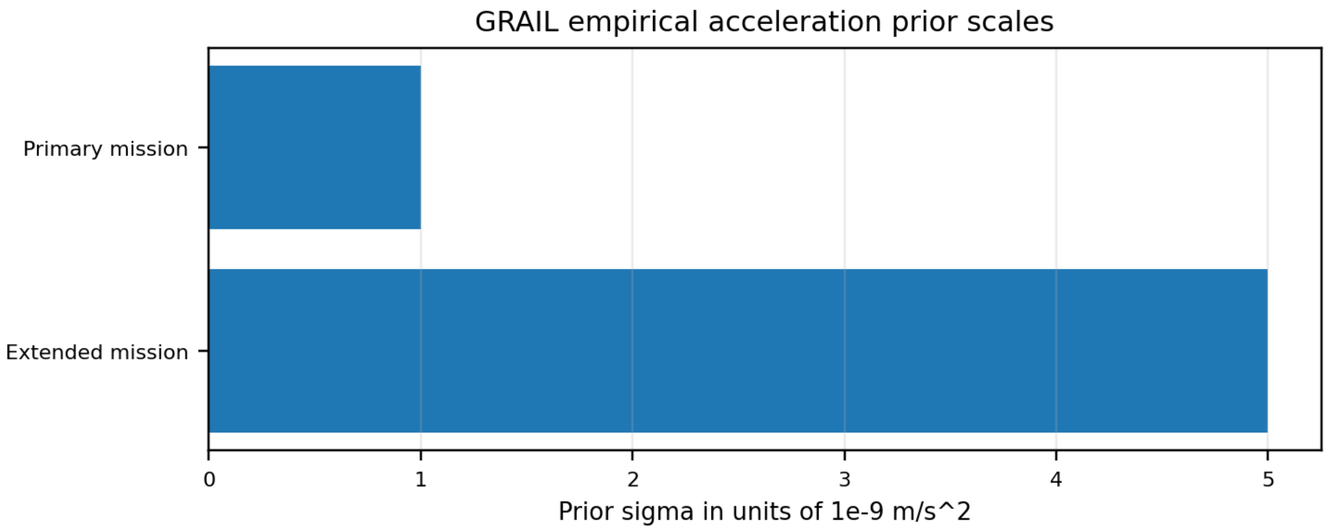


**Figure 11. GRAIL empirical acceleration prior scales.** Bars reproduce the published prior scales. They show acceleration freedom allowed by the solution, not a measured unexplained acceleration.

## 7.4 Common-mode and differential sensitivity

The Ka-band link measures relative motion. Nearly identical accelerations on both spacecraft can cancel substantially, while differential response remains if wheel states, attitudes, masses, or geometry differ. Individual DSN tracking is essential for a common-mode component. The combination is powerful: a local Ka-band structural disturbance may be strong in inter-spacecraft range rate but weak in independent center-of-mass orbit estimates; a true translational acceleration must propagate consistently into both, subject to common-mode rejection.

## 7.5 Archival products needed

| Product | Role |
| --- | --- |
| Signed wheel speed telemetry | Constructs per-wheel angular momentum and reversal intervals. |
| Attitude quaternions | Transforms body-fixed axes into a common inertial frame. |
| Wheel geometry and inertia | Defines axes and converts speed to angular momentum. |
| Thruster records | Masks or estimates momentum-unloading impulses. |
| Ka-band range and range rate | High-sensitivity differential observable and known resonance control. |
| DSN range and Doppler | Individual-spacecraft and common-mode trajectory sensitivity. |
| Mass, center of mass, and phase-center data | Separates trajectory motion from reference-point motion. |
| Orbit and force-model products | Reconstructs baseline and empirical absorption. |

## 7.6 Recommended experiment

Begin from pre-fit or lightly constrained radiometric data. Exclude thruster events and settling windows from continuous-force estimation. Retain approximately 100 Hz resonance windows as a separate control. Solve first without free 1/rev accelerations or with strong regularization, add the signed inertial wheel regressor, and only then restore conventional empirical terms. Estimate a common normalized alpha_K across both spacecraft while retaining wheel-specific columns long enough to detect mislabeled axes or local artifacts.

```
y = h(x_orbit, attitude, reference points, media, clocks) + epsilon
x_dot = f_standard(x,t) + alpha_K q_K^I(t) + a_empirical(t) + a_maneuver(t)
```

### 7.7 Positive-result criteria

A compelling result would be continuous between maneuvers, coherent with signed inertial wheel state, reversed under wheel reversal, present consistently in independent DSN observables, absent from the purely mechanical resonance control, and predictive across different wheel combinations, attitudes, mission phases, and both spacecraft.

## 8. Mars Odyssey: regular momentum unloading and trajectory prediction

### 8.1 Navigation context

Mars Odyssey navigation literature identifies angular-momentum desaturation as the largest error source in certain predicted trajectories [R3]. Desaturation thrusters impart real Delta-v, alter orbital period, and must be represented in reconstruction and prediction.

### 8.2 Published cadence and magnitude

Representative desaturations produced approximately 3-4 mm/s. Cadence was twelve hours in February-March 2015 and twenty hours in April-May 2015. The orbital period was approximately 1.96 h [R3]. Twelve hours spans approximately 6.12 Odyssey revolutions, and twenty hours approximately 10.20; the schedule changed while the orbit remained near two hours. This is direct evidence of an operational, not orbital, recurrence.

```
T_orbit,Odyssey approximately 1.96 h
T_desat = 12 h during Feb-Mar 2015
T_desat = 20 h during Apr-May 2015
Delta-v_desat approximately 3 to 4 mm/s
```

### 8.3 Telemetry-based scale factor

One prediction solution multiplied telemetry-derived desaturation Delta-v values by an empirical factor of 0.81 [R3]. The factor aggregates maneuver-reconstruction and background-model errors. It is a useful residual-search location but not specific evidence of a rotational force. A continuous K-Field term must be estimated separately from the known impulse.

### 8.4 Between-event test

Partition arcs into pre-event coast, burn, settling, and post-event coast. Model or mask burn and settling. Evaluate signed wheel angular momentum throughout coast intervals. The key question is whether Doppler residual curvature between unloads follows evolving inertial wheel state before the next thruster event.

| Interval | Dominant model | K-Field use |
|---|---|---|
| Pre-unload coast | Gravity, SRP, thermal recoil, continuous wheel state. | Primary continuous-force interval. |
| Desaturation burn | Finite thruster force and torque; wheel speed changes. | Estimate impulse; not primary signed-speed evidence. |
| Settling | Attitude and structural transients. | Mask or model with transient basis. |
| Post-unload coast | New wheel state with no active thruster. | Natural before-after test at nearly unchanged orbit geometry. |

### 8.5 Natural schedule-change experiment

A schedule-driven spectral line moves from twelve to twenty hours with commands. An orbit-locked term stays tied to the 1.96 h phase. An inertial K-Field term follows wheel-axis and field geometry and should not reset merely because the operations schedule changes.

# 9. RadioAstron: environmental force, wheel momentum, and unloading confounding

## 9.1 Mission geometry and solar force-torque coupling

RadioAstron operated in a highly elliptical Earth orbit with long inertial-pointing observations and large solar panels. Solar radiation pressure was important in both translation and torque. Force accelerated the center of mass; center-of-pressure offset generated torque that accumulated in reaction wheels. Wheel speeds were used to constrain SRP torque parameters [R4-R5].

This is the clearest common-cause example: an external force drives orbit perturbation while its associated torque drives wheel momentum. A regression that includes wheel state but omits solar geometry can produce a strong wheel coefficient even though the causal force is photon pressure. Force and torque parameters must be estimated jointly.

## 9.2 Momentum unloading

RadioAstron unloaded momentum approximately one or two times per day. Events lasted about two to three minutes and produced approximately 3-7 mm/s, commonly represented as impulses [R4-R5]. The integrated effect of unloading was comparable in navigation importance to the direct SRP perturbation. Prediction performance improved markedly when actual or regularized unloading histories were included.

## 9.3 Approximately 8.8-day residual

The space-VLBI delay-rate evaluation reported annual structure and a component near approximately 8.8 days, close to the relevant orbital period. The mean projected velocity residual before an additional correction was approximately 1.12 mm/s [R5]. The publication did not connect this term to signed wheel state; it was considered in orbit and correlator-model consistency. It is a methodological target, not a wheel-force identification.

| Quantity | Published value | Interpretation |
|---|---|---|
| Unload frequency | One or two events/day. | Operational cadence, not 12 h orbital fundamental. |
| Unload duration | Approximately 2-3 min. | Short compared with multi-day orbit. |
| Unload Delta-v | Approximately 3-7 mm/s. | Real external-actuator velocity change. |
| Residual modulation | Annual and approximately 8.8 d. | 8.8 d near orbit period; not tied to signed wheel state. |
| Mean projected velocity residual | Approximately 1.12 mm/s before extra correction. | Observation-model or orbit-consistency scale. |

## 9.4 Recommended separation model

```
a_total^I = a_gravity + a_SRP(attitude,Sun) + a_thermal + a_K(wheels,attitude,g_hat_K)
tau_total^I = tau_SRP(attitude,Sun,center_of_pressure) + tau_other
dH_RW^I/dt = -tau_body_control + commanded terms
```

A K-Field coefficient is credible only if it improves translation after the solar-force and solar-torque model independently explains the wheel-momentum history and unloading impulses.

# 10. GNSS precise orbit determination: the primary orbital-harmonic search domain

## 10.1 Why GNSS is uniquely relevant

GPS and GLONASS place their orbital fundamentals in or near the approximately twelve-hour band. GNSS precise orbit determination routinely estimates empirical harmonic accelerations and pseudo-stochastic corrections. This combination makes GNSS the direct domain for testing whether a twelve-hour term is a true orbital harmonic, an environmental-model error, or an estimator artifact. Galileo provides a control at approximately 14.08 h, and geosynchronous spacecraft provide a 2/rev route near twelve hours.

## 10.2 Empirical coefficient families

IGS Repro3 literature documents constants in body-related D, Y, and B directions; 1/rev sine and cosine terms such as B1S and B1C; 2/rev terms such as D2S and D2C; and in some solutions 4/rev or along-track harmonics [R7]. Naming varies, but the functional point is explicit basis functions at orbital harmonics.

| Family | Frequency | Likely absorbed effects | K-Field relevance |
|---|---|---|---|
| Constant D/Y/B | DC over interval. | Mean SRP bias, optical-property or attitude-force bias. | Can absorb nearly constant projection in short arc. |
| 1/rev sine/cosine | Orbital fundamental. | Fixed inertial projection, SRP error, thermal or attitude effects. | Direct GPS/GLONASS approximately 12 h absorption channel. |
| 2/rev sine/cosine | Second harmonic. | Quadratic or two-lobed geometry. | Direct GEO approximately 12 h absorption channel. |
| 4/rev terms | Fourth harmonic. | Higher-order force or attitude structure. | Control for repeated nonlinear geometry. |
| Pseudo-stochastic terms | Piecewise or impulsive. | Maneuvers and rapidly varying force errors. | Can absorb candidate if too flexible. |

## 10.3 Former twelve-hour spacing

Earlier CODE processing estimated certain pseudo-stochastic parameters every twelve hours. Revised scheduling around orbit midnight reduced orbit misclosures by approximately 10 percent for GPS and 15 percent for Galileo [R7]. This shows that estimator timing can create or remove structure without a twelve-hour physical oscillator. Alternate knot spacing is mandatory in a K-Field analysis.

## 10.4 Precision scale and solar models

For 2020, IGS Repro3 combined-orbit analysis reported mean orbit consistency of approximately 9 mm for GPS, 23 mm for GLONASS, and 15 mm for Galileo [R7]. Different SRP models can produce centimeter-scale radial differences, and some Galileo laser-ranging structures exceed 50 mm. These are the precision scales and nuisance structures against which a small periodic acceleration must be tested.

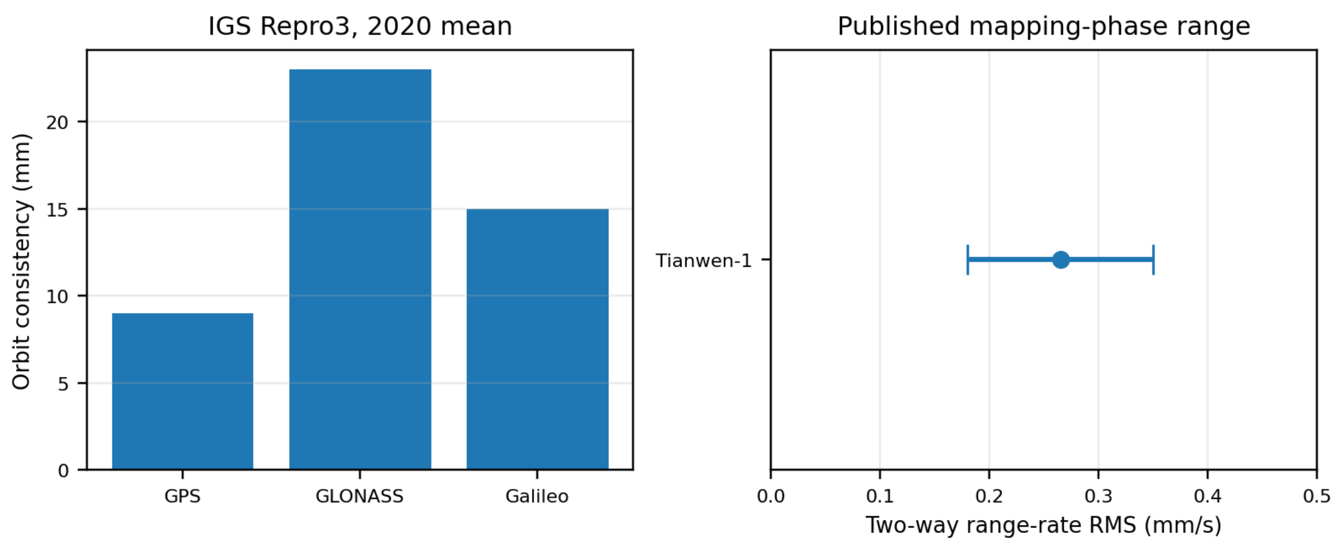


**Figure 12. Published precision scales for GNSS and Tianwen-1.** Left: mean 2020 IGS Repro3 orbit consistency for GPS, GLONASS, and Galileo. Right: Tianwen-1 mapping-phase two-way range-rate RMS. Panels retain native units.

## 10.5 Geosynchronous 2/rev

For GEO, the approximately twelve-hour search belongs in 2/rev coefficients; 1/rev belongs near a sidereal day. The fitted 2/rev phase must match the geometry predicted by the K-Field model rather than merely a spectral peak. Station keeping, unloading, solar-array articulation, yaw steering, eclipse seasons, Earth albedo, and thermal cycles are strong daily and twice-daily controls.

### 10.6 Constellation discriminators

| Discriminator | Orbital harmonic | Operational cadence | K-Field signed-wheel term |
|---|---|---|---|
| Period across constellations | Scales with each orbital period and selected harmonic. | Tracks policy, not orbit. | Projection follows orbit but amplitude and phase also follow wheel state and inertial field. |
| Phase reference | Closes with orbit or Sun-orbit geometry. | Resets at commands. | Closes with n_i^I and g_hat_K. |
| Wheel reversal | No required sign reversal. | Depends on sequence. | Odd component reverses. |
| Satellite coefficient | May follow optical properties and attitude law. | May follow operations and hardware. | Should normalize consistently under stated mass/inertia law. |

### 10.7 Recommended analysis

Reprocess multiple constellations with the same K-Field wheel regressor added before empirical harmonics. GPS and GLONASS test near-twelve-hour 1/rev; Galileo tests a different fundamental; GEO tests 2/rev. Coefficient stability across these transformations is stronger than any one spectral line. Direct wheel telemetry is the limiting asset for operational GNSS spacecraft.

## 11. Tianwen-1: orbit degradation from wheel off-loading and aggregate empirical acceleration

### 11.1 Published result

Tianwen-1 studies identify wheel off-loading as an important orbit-error source. Kong and coauthors report relay-orbit position accuracy near 150 m and remote-sensing-orbit accuracy near 700 m in the examined context [R8]. Liu and coauthors found one-day arcs inadequate for wheel off-loading and several unknown forces, while three-day arcs allowed an empirical acceleration to aggregate their effect [R9].

### 11.2 Residual scale and arc length

Three-day mapping-phase solutions reported two-way range-rate RMS of approximately 0.18-0.35 mm/s [R9]. This is a post-fit line-of-sight velocity scale, not a direct acceleration. Longer arcs improve geometry and support a smooth empirical term, but can correlate the candidate with initial state and nuisance accelerations.

| Choice | Benefit | Risk |
|---|---|---|
| One-day arc | Limits long-term drift and correlations. | May be too short; state reset absorbs low frequency. |
| Three-day arc | Improves geometry and allows aggregate acceleration. | Aggregate term may absorb K-Field with known force errors. |
| Explicit unload reconstruction | Represents real maneuver impulse. | Requires thrust, timing, attitude, mass. |
| Signed wheel fit | Directly tests rotational state. | Requires synchronized telemetry and nuisance basis. |

### 11.3 Recommended reanalysis

Retain multi-day geometry but replace the aggregate term in stages: estimate unloading impulses, fit SRP and thermal terms, add signed inertial wheel predictor, then restore a restricted empirical acceleration and measure absorption. Compare relay and remote-sensing phases for attitude and orbit diversity.

## 12. BepiColombo: simulated sensitivity to twice-daily desaturation

### 12.1 Study purpose and signatures

Alessi and coauthors studied desaturation effects on BepiColombo precise orbit determination [R10]. Maneuvers were solve-for quantities because they interrupt dynamical continuity. A baseline of two maneuvers per day produced meter-scale semimajor-axis changes, range signatures near 100 m, and range-rate signatures of some centimeters per second. These are known simulated impulse effects, not an unexplained twice-daily acceleration.

 

### 12.2 Control-case use

BepiColombo supplies a forward-model control for a twice-daily impulse train. An impulse train generates cadence harmonics and sharp time-domain transitions. A continuous K-Field term produces smooth inter-event curvature and signed-wheel dependence.

```
a_impulse(t) = sum_k Delta-v_k delta(t - t_k)
a_continuous(t) = alpha_K q_K^I(t)
```

## 13. Cross-mission synthesis

### 13.1 What the literature establishes

Reaction-wheel systems enter tracking and navigation through several pathways. Wheel-frequency disturbances contaminate observations. Momentum unloading produces real trajectory changes at several millimeters per second per event. Environmental torque makes wheel momentum a proxy for external force geometry. Orbit estimators use empirical terms at frequencies where an inertially directed acceleration projects.

Compensation is regular and mission critical: GRAIL deleted resonance intervals and fitted empirical accelerations; Odyssey scaled and scheduled unloading impulses; RadioAstron modeled hundreds of unloads and SRP torque; GNSS centers estimated orbital harmonics and pseudo-stochastic terms; Tianwen used longer arcs and aggregate acceleration; BepiColombo simulations treated desaturation as solve-for dynamics.

### 13.2 What it does not establish

No reviewed source isolates persistent center-of-mass acceleration from steady wheel rotation. No source reports a coefficient odd in signed wheel speed, reversed with wheel direction, following a fixed inertial field vector, and remaining after thruster, environmental, and structural modeling. The absence is primarily a model-definition absence: mission analyses were designed for navigation or instrument performance, not this test.

| Case | Wheel enters observable? | Real trajectory change? | Orbital term? | Signed inertial test? |
|---|---|---|---|---|
| GRAIL 100 Hz | Direct speed correlation. | Not demonstrated; likely reference-point/structural. | No, high-frequency speed-localized. | No. |
| GRAIL empirical | Possible absorption. | Could include real unmodeled force. | Constant and 1/rev. | No. |
| Odyssey unload | Wheel management triggers thrusters. | Yes, 3-4 mm/s. | 12 h is operational. | No. |
| RadioAstron | Wheel traces torque; unloads external. | Yes, SRP and 3-7 mm/s unloads. | Approximately 8.8 d near orbit. | No. |
| GNSS empirical | Public wheel telemetry usually absent. | Empirical terms may contain real force. | 1/rev, 2/rev, higher. | No. |
| Tianwen aggregate | Off-loading included. | Yes for unload; aggregate split unknown. | Not isolated. | No. |
| Bepi simulation | Desaturation explicitly modeled. | Yes in simulation. | Twice-daily event train. | No. |

### 13.3 Frequency-domain synthesis

| Band | Known terms | Candidate K-Field appearance | Control |
|---|---|---|---|
| Wheel frequency | Imbalance, bearings, motor ripple, cogging. | Not primary smooth band except modulation/resonance. | Wheel-angle lines and transfer functions. |
| Control bandwidth | Wheel acceleration, bus torque, loop transients. | Modulation during speed changes. | Motor current, d omega/dt, attitude error. |
| 1/rev | SRP, thermal, fixed inertial projection, gravity error. | Linear inertial-direction projection. | Sun geometry and multi-orbit phase closure. |
| 2/rev | Quadratic geometry, two-lobed force, estimator basis. | Possible nonlinear/symmetric projection; GEO near 12 h. | Derived phase and constellation periods. |
| Operational cadence | Unloading, station keeping, scheduled activity. | Indirectly through changed wheel state. | Thruster and command telemetry. |
| Estimator knots | Pseudo-stochastic terms and state resets. | Candidate may be removed or aliased. | Shift knots and arc lengths. |

 

### 13.4 Approximately twelve-hour conclusion

The approximately twelve-hour band is a true 1/rev orbital band for GPS and GLONASS, a 2/rev orbital band for geosynchronous spacecraft, and an operational or estimation interval in Odyssey, BepiColombo study assumptions, and former CODE spacing. Only the first two are orbital-harmonic interpretations.

### 13.5 Archival ranking

| Rank | Data set | Strength | Limitation |
|---|---|---|---|
| 1 | GRAIL | Public signed wheel speeds, attitude, thrusters, inter-spacecraft ranging, and individual DSN tracking; known resonance control. | Short lunar period, common-mode rejection, empirical absorption. |
| 2 | GNSS with wheel telemetry | Near-12 h fundamentals, long spans, multiple constellations, precise tracking. | Detailed wheel telemetry generally not public. |
| 3 | Mars Odyssey | Cadence changes, mm/s unloads, mature navigation. | Published association dominated by known thruster impulse. |
| 4 | RadioAstron | Strong force-torque coupling, many unloads, VLBI diagnostics. | SRP and wheel state highly correlated. |
| 5 | Tianwen-1 | Published precision and recognized off-loading problem. | Aggregate empirical treatment and limited telemetry. |
| 6 | BepiColombo | Defined simulated maneuver sensitivity. | Simulation rather than residual discovery. |

## 14. Decisive K-Field regression and signal-separation protocol

### 14.1 State reconstruction

For every wheel, reconstruct signed angular velocity, inertia, body-fixed axis, and spacecraft attitude. Transform the axis into a common inertial frame at every sample. Preserve a documented sign convention from telemetry channel to physical axis; a sign error can imitate failed reversal parity.

```
n_i^I(t) = R_I<-B(t) n_i,B
H_i^I(t) = I_i omega_i(t) n_i^I(t)
```

### 14.2 Candidate design columns

Construct three inertial components of the summed lateral proxy, then project into line of sight and RTN. Use the locked field direction and geometry rather than choosing a direction after inspecting residuals.

```
q_K^I(t) = sum_i I_i omega_i(t) [n_i^I(t) x g_hat_K]
q_K,LOS = l_hat^I dot q_K^I
q_K,R = r_hat^I dot q_K^I
q_K,T = t_hat^I dot q_K^I
q_K,N = n_hat^I dot q_K^I
```

### 14.3 Mandatory nuisance columns

| Family | Construction | Purpose |
|---|---|---|
| Wheel imbalance | omega_i^2 cos(theta_i), omega_i^2 sin(theta_i), identified harmonics. | Models even-speed wheel-angle disturbance. |
| Wheel acceleration | I_i d omega_i/dt and motor current. | Captures torque and control transients. |
| Structural modes | Band-limited modes or measured transfer functions. | Separates reference-point and line-of-sight vibration. |
| Momentum unloading | Finite-burn thrust or per-event Delta-v. | Represents real external impulse. |
| Solar radiation pressure | Surface or box-wing force using attitude and Sun. | Removes direct environmental acceleration correlated with wheel torque. |
| Solar torque | Center-of-pressure torque fitted to wheel momentum. | Uses wheel telemetry without assigning translation to wheel. |
| Thermal recoil | Temperatures, power, eclipse history, delayed basis. | Models illumination-dependent recoil. |
| Empirical orbit terms | Restricted constant, 1/rev, 2/rev, pseudo-stochastic basis. | Measures competition and absorption. |
| Reference-point motion | Attitude and elastic displacement of phase centers. | Separates path-length change from center-of-mass motion. |

## 14.4 Full model

```
a_observed^I(t) = a_standard^I(t)
+ alpha_K q_K^I(t)
+ a_omega2^I(t) + a_dotomega^I(t)
+ a_SRP^I(t) + a_thermal^I(t)
+ sum_k Delta-v_k delta(t-t_k)
+ a_empirical^I(t)
```

Acceleration enters the dynamic state equations and observations are functions of the integrated state. Use variational equations or equivalent sensitivities rather than fitting acceleration directly to already filtered residuals whenever possible.

## 14.5 Parity and inertial phase

The minimal K-Field term is odd in signed speed. Match positive and negative wheel intervals at similar inertial orientation and environmental conditions. Retain omega, omega squared, and d omega/dt simultaneously. Compute field-relative phase from attitude and locked g_hat_K. Body-fixed artifacts follow spacecraft coordinates, Sun-driven forces follow illumination coordinates, and the K-Field term follows inertial geometry.

```
q_K(-omega) = -q_K(omega)
q_imbalance_envelope(-omega) = q_imbalance_envelope(omega)
```

## 14.6 Maneuver exclusion and harmonic test

Model or mask thruster events with guard intervals based on burn duration, settling, thermal response, and residual autocorrelation. Do not fit one generic twelve-hour sinusoid to every spacecraft. Generate the predictor from wheel and attitude telemetry, then examine its orbital projection: GPS/GLONASS near-12 h 1/rev, GEO near-12 h 2/rev, GRAIL 110 min, Odyssey 1.96 h.

## 14.7 Estimator-absorption experiment

| Run | Model | Purpose |
|---|---|---|
| A | Deterministic standard model plus maneuvers. | Baseline residual structure. |
| B | A plus signed K-Field regressor. | Direct candidate coefficient. |
| C | A plus conventional 1/rev and 2/rev terms. | Standard absorption without K term. |
| D | A plus K term plus empirical harmonics. | Tests survival under competition. |
| E | D with shifted knots and alternate arcs. | Tests estimator architecture dependence. |
| F | D on independent spacecraft, attitudes, and time spans. | Tests transportability. |

## 14.8 Cross-validation

Use physically distinct holdouts rather than random sample removal: complete wheel reversals, attitude modes, seasons, maneuver densities, spacecraft, and constellations. Random holdouts can overstate success in autocorrelated residuals.

| Split | Question |
|---|---|
| Wheel reversal | Does the fitted coefficient predict sign reversal? |
| Attitude mode | Does inertial geometry transfer beyond one body orientation? |
| Season or Sun angle | Is the coefficient independent of SRP and thermal geometry? |
| Maneuver density | Does it persist when unload frequency changes? |
| Spacecraft | Does one normalization transfer to another vehicle? |
| Constellation | Does the model transform correctly among orbital periods and harmonics? |

## 14.9 Required outputs

Report fitted coefficient and units, covariance, parameter correlations, pre/post residual spectra, time-domain predictive error, sensitivity to arc and knot choices, and every predeclared control. Do not report only a dimensionless significance value.

 

# 15. Mission-specific experimental programs

## 15.1 GRAIL archival program

| Stage | Action | Acceptance |
|---|---|---|
| 1 | Reconstruct signed wheel angular momentum and inertial axes. | Axis signs, timestamps, and units pass continuity and maneuver checks. |
| 2 | Reproduce baseline and identify approximately 100 Hz intervals. | Residual and event behavior agree with published processing. |
| 3 | Estimate K term on DSN data with thruster and resonance masks. | Stable across range and Doppler; not event-driven. |
| 4 | Propagate individual orbits into predicted Ka-band relative motion. | Differential effect matches independent trajectory prediction. |
| 5 | Restore empirical accelerations and vary regularization. | Survival or quantitative absorption is mapped. |
| 6 | Transfer between spacecraft and mission phases. | Normalized coefficient agrees within uncertainty. |

## 15.2 GNSS and GEO orbital-harmonic program

| Stage | Action | Acceptance |
|---|---|---|
| 1 | Acquire wheel telemetry, attitude, bus geometry, and precise tracking. | Synchronized signed wheel histories at adequate cadence. |
| 2 | Generate inertial and RTN K regressors. | 1/rev and 2/rev content follows each orbit. |
| 3 | Reprocess without free harmonics, then add K term. | Improves multiple observables without phase tuning. |
| 4 | Compete with SRP and empirical harmonics. | Coefficient remains stable. |
| 5 | Compare GPS, GLONASS, Galileo, and GEO. | Near-12 h appears as correct 1/rev or 2/rev transformation, not universal clock sinusoid. |

## 15.3 Mars, RadioAstron, and purpose-built programs

Use Odyssey and Tianwen-1 as complementary Mars cases, separating coast acceleration from unloading impulse across mapping and relay phases. For RadioAstron, jointly fit solar force, solar torque, wheel momentum, unloading impulses, and K-Field term; the K coefficient must improve translation without degrading the independent torque fit.

A purpose-built spacecraft should use counter-rotating wheels whose total angular momentum, kinetic energy, mass, and speed can be varied independently; perform repeated reversals, inertial attitude sweeps, and long thruster-free coasts; and use multiple independent translation sensors. Equal-energy unequal-mass wheel states can determine whether coupling follows angular momentum, angular speed, kinetic energy, mass distribution, or another quantity. Combining a near-twelve-hour MEO with GEO or a short orbit separates orbital and daily cycles.

# 16. Conclusions

Known literature contains multiple periodic and recurrent drift terms connected to reaction-wheel systems, but their pathways differ. Wheel harmonics create structural vibration, pointing jitter, and tracking reference-point contamination. Momentum unloading creates real center-of-mass velocity changes at millimeter-per-second scale. Orbit-determination systems introduce empirical periodic accelerations and pseudo-stochastic corrections that can absorb real or modeled force errors.

GRAIL provides the strongest direct published wheel-speed correlation in precision tracking: high-frequency Ka-band range-rate disturbances near an approximately 100 Hz wheel-speed region. The signature is most consistent with mechanical resonance because it is abrupt, narrow in speed, and short in duration. GRAIL is also the strongest accessible low-frequency test because its orbit solutions used 10^-9 to 5 x 10^-9 m/s^2 empirical acceleration priors and its archive includes the telemetry required for a signed inertial predictor.

Mars Odyssey and RadioAstron show that momentum unloading can dominate prediction. Odyssey produced approximately 3-4 mm/s events and changed from twelve-hour to twenty-hour cadence. RadioAstron produced approximately 3-7 mm/s one or two times per day while SRP drove both orbit and wheel momentum. These known effects are indispensable controls.

 

The approximately twelve-hour effect is relevant only when phase has a justified orbital source. GPS and GLONASS provide near-twelve-hour 1/rev fundamentals. GEO provides a 23 h 56 min 4 s fundamental and an 11 h 58 min 2 s 2/rev harmonic. GRAIL, Odyssey, Galileo, and RadioAstron occupy different bands. Commanded events and estimator knots at twelve hours are not orbital harmonics.

The decisive K-Field test includes wheel inertia, signed speed, axis geometry, attitude, locked inertial field direction, wheel-angle harmonics, wheel acceleration, solar force and torque, thermal recoil, thruster impulses, reference-point motion, and estimator absorption. The candidate must be continuous between unloads, odd under reversal, inertially phase-locked, and stable across spacecraft and orbit geometries.

No reviewed publication performs that separation. The literature nevertheless identifies where the signal can reside and how processing can remove it. GRAIL is the highest-priority public archival experiment. Telemetry-enabled GNSS and GEO are the highest-priority approximately twelve-hour orbital-harmonic experiment.

**Final statement**

Do not search for an unqualified twelve-hour residual. Search for the predeclared signed-wheel inertial predictor, then verify that its orbital projection occurs at the correct fundamental or harmonic while surviving maneuver, vibration, environmental, and estimator controls.

 

# Appendix A. Derivations and dimensional checks

## A.1 Center-of-mass equation

For components m_j at positions r_j, the center of mass is R = M^-1 sum m_j r_j. Summing Newtonian equations cancels internal action-reaction pairs and leaves only external forces.

```
R = (1/M) sum_j m_j r_j
M d^2R/dt^2 = sum_j F_j,external + sum_j F_j,internal
sum_j F_j,internal = 0
M d^2R/dt^2 = sum_j F_j,external
```

Reaction-wheel bearing forces and motor torques are internal in the conventional closed model. Thruster exhaust, photons, magnetic exchange, atmosphere, and the proposed K-Field coupling are external channels that must be represented explicitly.

## A.2 Fixed inertial vector projected into RTN

Choose fixed orthonormal p and q vectors in the orbit plane and h = p x q. The rotating radial and transverse axes give the following dot products for a fixed inertial vector a^I = a_p p + a_q q + a_h h.

```
a_R = a_p cos u + a_q sin u
a_T = -a_p sin u + a_q cos u
a_N = a_h
```

Slow plane precession, eccentricity, attitude changes, and field-amplitude modulation add sidebands and changing coefficients but preserve the basic 1/rev mapping.

## A.3 Second harmonic from quadratic geometry

```
s(u) = C cos(u - phi)
s^2(u) = (C^2/2) [1 + cos(2u - 2phi)]
```

The 2/rev phase is twice the underlying orientation phase. A free sine-cosine pair can match any phase; a physical model predicts this relation.

## A.4 Sinusoidal acceleration integration

```
a(t) = A cos(omega t)
v(t) = (A/omega) sin(omega t) + C_v
x(t) = -(A/omega^2) cos(omega t) + C_v t + C_x
v_amp = A/omega
x_amp = A/omega^2
```

At T = 43200 s and A = 10^-9 m/s^2, amplitudes are approximately 6.87549 x 10^-6 m/s and 4.72724 x 10^-2 m. At T = 6600 s, they are approximately 1.05042 x 10^-6 m/s and 1.10339 x 10^-3 m.

## A.5 Signed-speed parity

```
y_K(omega) = beta omega
y_K(-omega) = -beta omega
y_I(omega) = gamma omega^2
y_I(-omega) = gamma omega^2
```

A full imbalance signal also contains wheel-angle sine and cosine functions. Reversal changes phase progression but not the quadratic envelope.

## A.6 Impulse train versus continuous acceleration

```
a_impulse(t) = sum_k Delta-v_k delta(t - t_k)
v(t_k^+) - v(t_k^-) = Delta-v_k
a_continuous(t) = alpha_K q_K(t)
```

An impulse train creates cadence harmonics but retains sharp event-time structure; a continuous candidate creates smooth evolution between events.

### A.7 Geosynchronous and GPS period calculation

```
T_sidereal = 23 h + 56 min + 4 s = 23.934444444 h
T_GEO,2/rev = T_sidereal / 2 = 11.967222222 h = 11 h 58 min 2 s
```

GPS nominal motion is approximately two revolutions per sidereal day, so GPS 1/rev and GEO 2/rev have nearly equal numerical periods but different harmonic labels.

# Appendix B. K-Field Base Cell geometric ledger

### B.1 Edge vectors

| Vector | x / L_K | y / L_K | z / L_K | Magnitude / L_K |
|---|---|---|---|---|
| a | 0.020797058700 | 0.000000000000 | 0.000000000000 | 0.0207970587 |
| b | 0.003856816120 | 0.036289833047 | 0.000000000000 | 0.0364942052 |
| c | 0.008182284384 | 0.030762382380 | 0.037657147177 | 0.0493085660 |

### B.2 Gram matrix

| Row | a column | b column | c column | Unit |
|---|---|---|---|---|
| a | 4.325176505712456e-04 | 8.021043124313843e-05 | 1.701674486329727e-04 | L_K^2 |
| b | 8.021043124313843e-05 | 1.331827013179707e-03 | 1.147919287031111e-03 | L_K^2 |
| c | 1.701674486329727e-04 | 1.147919287031111e-03 | 2.431334680976355e-03 | L_K^2 |

Diagonal entries are squared edge lengths; off-diagonal entries reproduce pairwise angles. det(G) equals V squared.

### B.3 Nodes

| Node | x / L_K | y / L_K | z / L_K | Definition |
|---|---|---|---|---|
| O | 0.000000000000 | 0.000000000000 | 0.000000000000 | origin |
| A | 0.020797058700 | 0.000000000000 | 0.000000000000 | a |
| B | 0.003856816120 | 0.036289833047 | 0.000000000000 | b |
| C | 0.008182284384 | 0.030762382380 | 0.037657147177 | c |
| AB | 0.024653874820 | 0.036289833047 | 0.000000000000 | a + b |
| AC | 0.028979343084 | 0.030762382380 | 0.037657147177 | a + c |
| BC | 0.012039100504 | 0.067052215428 | 0.037657147177 | b + c |
| ABC | 0.032836159204 | 0.067052215428 | 0.037657147177 | a + b + c |

### B.4 Face areas and membership

| Face | Origin-face vertices | Opposite-face vertices | Area |
|---|---|---|---|
| a-b | O, A, AB, B | C, AC, ABC, BC | 0.000754721788097 L_K^2 |
| a-c | O, A, AC, C | B, AB, ABC, BC | 0.001011255755679 L_K^2 |
| b-c | O, B, BC, C | A, AB, ABC, AC | 0.001385784440910 L_K^2 |

## B.5 Plane equations

| Plane | Spanning vectors | Unit normal (x, y, z) | Origin face | Opposite-face offset |
|---|---|---|---|---|
| P_ab | a, b | (0.000000000000, 0.000000000000, 1.000000000000) | n_ab dot x = 0 | 0.037657147177 |
| P_ac | a, c | (0.000000000000, -0.774440981837, 0.632646161492) | n_ac dot x = 0 | -0.028104333936 |
| P_bc | b, c | (0.986135753693, -0.104804678115, -0.128655566277) | n_bc dot x = 0 | 0.020508723156 |

## B.6 Volume checks

```
V = | a dot (b x c) |
V = 2.842066945211217e-05 L_K^3
det(G) = V^2
```

The calculated face areas are A_ab = 0.000754721788097 L_K^2, A_ac = 0.001011255755679 L_K^2, and A_bc = 0.001385784440910 L_K^2. The volume is 2.842066945211217e-05 L_K^3.

## B.7 Use in spacecraft regression

Mission wheel axes are not assumed to coincide with a, b, or c. Attitude maps hardware axes into inertial space, and the fixed K-Field operator maps those axes relative to the locked field direction. The Base Cell is not reoriented after observing the data.

# Appendix C. Orbital-period and harmonic ledger

| System or term | Period (h) | Classification | 12 h relevance | Source |
|---|---|---|---|---|
| GRAIL 1/rev | 1.833333333 | orbital fundamental | No direct 12 h identification | R1 |
| Mars Odyssey 1/rev | 1.960000000 | orbital fundamental | No direct 12 h identification | R3 |
| GLONASS 1/rev | 11.262222222 | orbital fundamental | Direct orbital relevance | R13 |
| GPS 1/rev | 11.967222222 | orbital fundamental | Direct orbital relevance | R11 |
| GEO 2/rev | 11.967222222 | second orbital harmonic | Direct orbital relevance | R12 |
| Galileo 1/rev | 14.079166667 | orbital fundamental | No direct 12 h identification | R14 |
| GEO 1/rev | 23.934444444 | orbital fundamental | Parent fundamental of GEO 2/rev | R12 |
| RadioAstron mean 1/rev | 211.200000000 | approximate orbital fundamental | No direct 12 h identification | R5 |

| Term | Period (h) | Cycles/day | Angular frequency (rad/s) | Role |
|---|---|---|---|---|
| GRAIL 1/rev | 1.833333333 | 13.090909091 | 9.519977738151e-04 | orbital fundamental |
| Mars Odyssey 1/rev | 1.960000000 | 12.244897959 | 8.904741081604e-04 | orbital fundamental |
| GLONASS 1/rev | 11.262222222 | 2.131018153 | 1.549720132986e-04 | orbital fundamental |
| GPS 1/rev | 11.967222222 | 2.005477926 | 1.458424703398e-04 | orbital fundamental |
| GEO 2/rev | 11.967222222 | 2.005477926 | 1.458424703398e-04 | second orbital harmonic |
| Galileo 1/rev | 14.079166667 | 1.704646345 | 1.239653804317e-04 | orbital fundamental |
| GEO 1/rev | 23.934444444 | 1.002738963 | 7.292123516990e-05 | orbital fundamental |
| RadioAstron mean 1/rev | 211.200000000 | 0.113636364 | 8.263869564367e-06 | approximate orbital fundamental |

| Case | Numerical period | Correct classification |
|---|---|---|
| GPS | Approximately 11 h 58 min | 1/rev orbital fundamental. |
| GLONASS | 11 h 15 min 44 s | 1/rev orbital fundamental. |
| GEO | 11 h 58 min 2 s | 2/rev harmonic; fundamental is 23 h 56 min 4 s. |
| Mars Odyssey | 12 h in Feb-Mar 2015 | Commanded unloading cadence; orbit approximately 1.96 h. |
| BepiColombo baseline | Two desaturations/day | Study-defined maneuver cadence. |
| Former CODE | 12 h spacing | Estimator knot spacing. |

# Appendix D. Published metric ledger

| System | Metric | Value | Unit or qualifier | Source |
|---|---|---|---|---|
| GRAIL | Wheel-speed-associated KBRR disturbance | approximately 100 | Hz wheel frequency | R1 |
| GRAIL | Disturbance duration | tens | seconds | R1 |
| GRAIL | Orbit period | 110 | minutes | R1 |
| GRAIL | Primary empirical acceleration prior sigma | 1e-9 | m/s^2 | R1 |
| GRAIL | Extended empirical acceleration prior sigma | 5e-9 | m/s^2 | R1 |
| GRAIL | Empirical acceleration update spacing | quarter orbit | schedule | R1 |
| Mars Odyssey | Orbit period | 1.96 | hours | R3 |
| Mars Odyssey | Typical desaturation Delta-v | 3-4 | mm/s | R3 |
| Mars Odyssey | Desaturation cadence, Feb-Mar 2015 | 12 | hours | R3 |
| Mars Odyssey | Desaturation cadence, Apr-May 2015 | 20 | hours | R3 |
| Mars Odyssey | Example telemetry Delta-v scale factor | 0.81 | dimensionless | R3 |
| RadioAstron | Desaturation cadence | 1-2 | events/day | R4 |
| RadioAstron | Desaturation duration | 2-3 | minutes | R4 |
| RadioAstron | Desaturation Delta-v | 3-7 | mm/s | R4-R5 |
| RadioAstron | Residual modulation near orbit period | approximately 8.8 | days | R5 |
| RadioAstron | Mean projected velocity residual before extra correction | 1.12 | mm/s | R5 |
| GNSS Repro3 | Mean orbit consistency GPS, 2020 | 9 | mm | R7 |
| GNSS Repro3 | Mean orbit consistency GLONASS, 2020 | 23 | mm | R7 |
| GNSS Repro3 | Mean orbit consistency Galileo, 2020 | 15 | mm | R7 |
| GNSS CODE | Former pseudo-stochastic spacing | 12 | hours | R7 |
| GNSS CODE | Misclosure reduction after rescheduling, GPS | 10 | percent | R7 |
| GNSS CODE | Misclosure reduction after rescheduling, Galileo | 15 | percent | R7 |
| Tianwen-1 | Relay-orbit position accuracy | approximately 150 | m | R8 |
| Tianwen-1 | Remote-sensing-orbit position accuracy | approximately 700 | m | R8 |
| Tianwen-1 | Two-way range-rate RMS | 0.18-0.35 | mm/s | R9 |
| BepiColombo study | Desaturation baseline | 2 | events/day | R10 |
| BepiColombo study | Semimajor-axis effect | meters | order of magnitude | R10 |
| BepiColombo study | Range effect | approximately 100 | m | R10 |
| BepiColombo study | Range-rate effect | some | cm/s | R10 |
| GEO | Orbital fundamental | 23 h 56 min 4 s | sidereal day | R12 |
| GEO | Second harmonic | 11 h 58 min 2 s | 2/rev | derived from R12 |
| GPS | Orbital fundamental | approximately 11 h 58 min | 1/rev | R11 |
| GLONASS | Orbital fundamental | 11 h 15 min 44 s | 1/rev | R13 |
| Galileo | Orbital fundamental | 14 h 4 min 45 s | 1/rev | R14 |

Rows marked derived are arithmetic transformations of published inputs. Source magnitude categories are preserved rather than converted into false precision.

# Appendix E. Figure and table data provenance

| Figure | Subject | Inputs | Provenance |
|---|---|---|---|
| 1 | Odd/even parity | y = omega and y = omega^2. | Derived equation |
| 2 | Unloading Delta-v | Odyssey 3-4; RadioAstron 3-7 mm/s. | R3-R5 |
| 3 | Orbital period map | Published periods and derived GEO 2/rev. | R1, R3, R5, R11-R14 |
| 4 | RTN projection | Printed components 0.8, 0.6, 0.3. | Derived equation |
| 5 | Unloading cadence | Odyssey 12 and 20 h; RadioAstron 1-2/day; Bepi 2/day. | R3, R4, R10 |
| 6 | Base Cell 3D | Locked lengths, angles, calculated nodes. | K-Field constants |
| 7 | Base Cell planes | Locked lengths, angles, areas. | K-Field constants |
| 8 | Orientation factor | \|n x g_hat_K\| sweep. | Locked field direction |
| 9 | Displacement scale | A/omega^2 at 10^-9 m/s^2. | Derived equation |
| 10 | Velocity scale | A/omega at 10^-9 m/s^2. | Derived equation |
| 11 | GRAIL priors | 1 and 5 x 10^-9 m/s^2. | R1 |
| 12 | Precision scales | GNSS consistency and Tianwen RMS. | R7, R9 |

## E.1 Defensive layout controls

| Control | Implementation |
|---|---|
| Fonts | PDF text uses only Helvetica, Helvetica-Bold, Times-Roman, and Courier. |
| Table widths | All column widths are fractions of available width DW = page width minus margins. |
| Cell wrapping | Every textual table cell is a Paragraph. |
| Page splitting | LongTable uses repeatRows = 1 and splitByRow = 1. |
| Figures | Aspect-ratio dimensions are constrained to available text width and maximum height. |
| Navigation | Headings generate bookmarks and a multi-pass table of contents. |

# Appendix F. Source ledger and references

Source identifiers in the body and JSON resolve to the entries below. URLs are included for archival traceability. Proposed K-Field equations are model terms, not claims taken from these mission sources.


[R1] Goossens, S.; Sabaka, T. J.; Genova, A.; Mazarico, E.; Nicholas, J. B.; Neumann, G. A.; Lemoine, F. G.; Zuber, M. T.; Smith, D. E. (2020). High-Resolution Gravity Field Models from GRAIL Data and Implications for Models of the Density Structure of the Moon's Crust. Journal of Geophysical Research: Planets, 125(2). DOI: 10.1029/2019JE006086. URL: https://doi.org/10.1029/2019JE006086. Role: GRAIL wheel-speed-correlated KBRR disturbance and empirical acceleration parameterization.

[R2] Jet Propulsion Laboratory / NASA Planetary Data System (2016). GRAIL Data Product Software Interface Specification, Version 1.8, JPL D-76383. NASA PDS Geosciences Node. URL: https://pds-geosciences.wustl.edu/grail/grail-l-lgrs-2-edr-v1/grail_0001/document/dpsis.htm. Role: Official archive definitions for wheel speed, attitude, thruster, KBRR, and DSN products.

[R3] Esposito, P.; Jefferson, D. C.; Lee, J. (2015). Odyssey Mars Orbiter - Thirteen Years of On-Orbit Navigation. 25th International Symposium on Space Flight Dynamics. URL: https://issfd.org/2015/files/downloads/papers/020_Esposito.pdf. Role: Mars Odyssey orbit period, desaturation cadence, typical Delta-v, and 0.81 scale factor.

[R4] Zakhvatkin, M. V.; Ponomarev, D. A.; Stepanyants, V. A.; Tuchin, A. G.; Zaslavskiy, G. S. (2014). Determination and Prediction of Orbital Parameters of the RadioAstron Mission. 24th International Symposium on Space Flight Dynamics. URL: https://issfd.org/ISSFD_2014/ISSFD24_Paper_S18-5_zakhvatkin.pdf. Role: RadioAstron wheel unloading frequency, Delta-v, impulse modeling, and SRP torque estimation.

[R5] Zakhvatkin, M. V.; et al. (2020). RadioAstron Orbit Determination and Evaluation of Its Results Using Correlation of Space-VLBI Observations. Advances in Space Research, 65(2), 798-812. DOI: 10.1016/j.asr.2019.10.012. URL: https://arxiv.org/abs/1812.01623. Role: Seven-year orbit assessment, unload effects, and delay-rate residual periodicities.

[R6] Dennehy, C. J. (2019). A Survey of Reaction Wheel Disturbance Modeling Approaches for Spacecraft Line-of-Sight Jitter Performance Analysis. European Space Mechanisms and Tribology Symposium. URL: https://esmats.eu/esmatspapers/pastpapers/pdfs/2019/dennehy.pdf. Role: Conventional wheel harmonics, imbalance, structural resonance, and jitter pathways.


 

[R7] Zajdel, R.; et al. (2023). Combination and SLR Validation of IGS Repro3 Orbits for ITRF2020. Journal of Geodesy, 97, 87. DOI: 10.1007/s00190-023-01777-3. URL: https://doi.org/10.1007/s00190-023-01777-3. Role: GNSS empirical harmonics, pseudo-stochastic parameters, former 12-hour spacing, and orbit consistency.

[R8] Kong, J.; Zhang, Y.; Chen, M.; Duan, J.; Li, C. (2024). The Effect of Wheel Off-Loading on the Orbit of Tianwen-1. Journal of Deep Space Exploration, 11(4), 414-420. DOI: 10.15982/j.issn.2096-9287.2024.20230010. URL: https://doi.org/10.15982/j.issn.2096-9287.2024.20230010. Role: Tianwen-1 orbit degradation attributed to wheel off-loading.

[R9] Liu, S.; Kong, J.; Cao, J.; Huang, H.; Man, H.; Yan, J.; Li, X. (2024). Precise Orbit Determination for Tianwen-1 During Mapping Phase. Astrodynamics, 8, 471-481. DOI: 10.1007/s42064-024-0226-4. URL: https://doi.org/10.1007/s42064-024-0226-4. Role: One-day versus three-day arcs, empirical acceleration aggregation, and range-rate residuals.

[R10] Alessi, E. M.; Cicalo, S.; Milani, A.; Tommei, G. (2012). Desaturation Manoeuvres and Precise Orbit Determination for the BepiColombo Mission. Monthly Notices of the Royal Astronomical Society, 423(3), 2270-2278. DOI: 10.1111/j.1365-2966.2012.21035.x. URL: https://doi.org/10.1111/j.1365-2966.2012.21035.x. Role: Twice-daily desaturation simulation and orbit/range effects.

[R11] United States Space Force / GPS.gov (2026). GPS Space Segment. Official GPS public information site. URL: https://www.gps.gov/space-segment. Role: GPS satellites circle Earth twice per sidereal day.

[R12] European Space Agency (2026). Types of Orbits. ESA Space Transportation. URL: https://www.esa.int/Enabling_Support/Space_Transportation/Types_of_orbits. Role: Geostationary period of 23 h 56 min 4 s.

[R13] GLONASS Information-Analytical Center (2026). About GLONASS. Official GLONASS information site. URL: https://glonass-iac.ru/en/about_glonass/. Role: GLONASS nominal period of 11 h 15 min 44 s.

[R14] European Space Agency (2026). ESA GNSS Book, TM-23, Volume I. ESA Navigation Support Office. URL: https://gssc.esa.int/navipedia/GNSS_Book/ESA_GNSS-Book_TM-23_Vol_I.pdf. Role: Galileo nominal period of 14 h 4 min 45 s.

[R15] Tapley, B. D.; Schutz, B. E.; Born, G. H. (2004). Statistical Orbit Determination. Elsevier Academic Press. Role: Estimation and observation-model reference.

[R16] Montenbruck, O.; Gill, E. (2000). Satellite Orbits: Models, Methods, and Applications. Springer. DOI: 10.1007/978-3-642-58351-3. URL: https://doi.org/10.1007/978-3-642-58351-3. Role: Orbital frames, perturbations, and precise orbit determination reference.

# Appendix G. Machine-readable companion schema

The companion JSON uses the same filename stem as the PDF. It contains exact Base Cell nodes and plane normals, period values used by figures, mission metrics, source ledger, figure ledger, layout audit, test equations, and the final PDF SHA-256 digest.

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{
"metadata": { ... },
"scope": { ... },
"central_findings": [ ... ],
"effect_classes": [ ... ],
"orbital_periods": [ ... ],
"mission_metrics": [ ... ],
"kfield_base_cell": { ... },
"kfield_test_model": { ... },
"figures": [ ... ],
"tables": { ... },
"sources": [ ... ],
"integrity": { ... }
}
```

 

# Appendix H. Glossary

| Term | Definition |
|---|---|
| Attitude | Spacecraft body orientation relative to inertial, orbital, Earth-fixed, Sun, or target frame. |
| Center of mass | Mass-weighted mean position of the complete system. |
| Desaturation | External-actuator removal of reaction-wheel angular momentum. |
| Doppler or range rate | Observable primarily sensitive to line-of-sight relative velocity plus propagation, clock, attitude, and reference-point effects. |
| Empirical acceleration | Estimated acceleration basis absorbing force-model errors without unique physical assignment. |
| Geostationary | Circular equatorial prograde geosynchronous orbit fixed in longitude. |
| Geosynchronous | Earth orbit with sidereal-day period; not necessarily circular or equatorial. |
| Harmonic | Integer multiple of a fundamental frequency; 2/rev completes two cycles/orbit. |
| ICRS | International Celestial Reference System used for inertial Cartesian orientation. |
| Impulse | Force integral; per unit mass a Delta-v. |
| Ka-band range rate | GRAIL inter-spacecraft separation-rate observable. |
| K-Field Base Cell | Locked parallelepiped generated by a, b, c in native units. |
| Line-of-sight jitter | Rapid pointing or reference-point motion projected along measurement direction. |
| Orbit determination | State and parameter estimation from tracking and equations of motion. |
| Pseudo-stochastic parameter | Piecewise parameter at selected epochs absorbing unmodeled dynamics. |
| Reaction wheel | Internal motor-driven flywheel for attitude control. |
| RTN frame | Radial, transverse/along-track, orbit-normal frame. |
| Signed wheel speed | Angular speed carrying direction under fixed physical axis convention. |
| Solar radiation pressure | Photon momentum-transfer force and torque. |

# Archival colophon

This self-contained reference was generated with Python and ReportLab. PDF text uses only the standard core fonts Helvetica, Helvetica-Bold, Times-Roman, and Courier. Long tables use explicit widths calculated from available page width, Paragraph-wrapped textual cells, and repeated header rows after page breaks.

Figures were generated programmatically from the numerical data and equations recorded in this report and companion JSON. Mission-specific plotted values trace to the source ledger. Base Cell figures use the locked edge magnitudes, angles, node coordinates, face areas, plane normals, and volume printed in Appendix B.

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